
Deep Active Learning for Precision Agriculture: A Computational Approach

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To God.

*To my parents,
Welison and Maria.*

In memoriam *of my maternal grandparents,
Nonato and Esmerinda.*

*To my paternal grandparents,
Doralice and Valderico.*

*To my beloved wife,
Thalia.*

To my friends.

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Abstract

Precision agriculture leverages remote sensing, autonomous vehicles, and real-time analytics to optimize inputs and enhance crop productivity across diverse environments. Integrating computer vision and machine learning further enables precise detection and targeted management of phytosanitary issues, improving efficiency and sustainability. Accurate weed species identification and classification are crucial for targeted interventions in precision agriculture; however, manually annotating large image datasets remains a significant bottleneck in this process. Recent deep active learning (DAL) methods have shown promise in reducing annotation effort while maintaining high performance in weed detection and classification tasks. In this study, we employ a dataset of 8,086 field-acquired images and start our pipeline with 100 labeled samples, iteratively querying batches of 10, 50, and 100 images using margin-based uncertainty sampling. We fine-tune a ResNet50 backbone, selected for its strong generalization ability, and rigorously evaluate its performance using F1-score, precision, and recall on an independent test set. Using MarginSampling with 100-image queries, the DAL strategy achieved an F1-score of 89.14%, close to the 91.34% obtained with the full dataset, while reducing the labeling requirements by approximately 86%. These results reinforce the viability of DAL as a scalable and cost-effective solution for real-time weed classification in precision agriculture. Future work will extend this framework to multimodal models using images and textual metadata, and this abstract represents the first step in a broader PhD investigation on DAL in precision agriculture.

Keywords: Deep Active Learning, Weed Classification, Precision Agriculture

Resumo

A agricultura de precisão utiliza sensoriamento remoto, veículos autônomos e análises em tempo real para otimizar insumos e aumentar a produtividade das culturas em diversos ambientes. A integração da visão computacional e do aprendizado de máquina permite ainda mais a detecção precisa e o gerenciamento direcionado de problemas fitossanitários, melhorando a eficiência e a sustentabilidade. A identificação e a classificação precisas de espécies de plantas daninhas são cruciais para intervenções direcionadas na agricultura de precisão; no entanto, a anotação manual de grandes conjuntos de dados de imagens continua sendo um gargalo significativo nesse processo. Métodos recentes de aprendizado ativo profundo (DAL) têm se mostrado promissores na redução do esforço de anotação, mantendo alto desempenho em tarefas de detecção e classificação de plantas daninhas. Neste estudo, empregamos um conjunto de dados de 8.086 imagens adquiridas em campo e iniciamos nosso pipeline com 100 amostras rotuladas, consultando iterativamente lotes de 10, 50 e 100 imagens usando amostragem de incerteza baseada em margem. Ajustamos um backbone ResNet50, selecionado por sua forte capacidade de generalização, e avaliamos rigorosamente seu desempenho usando pontuação F1, precisão e recall em um conjunto de teste independente. Utilizando MarginSampling com consultas de 100 imagens, a estratégia DAL alcançou uma pontuação F1 de 89,14%, próxima aos 91,34% obtidos com o conjunto de dados completo, reduzindo os requisitos de rotulagem em aproximadamente 86%. Esses resultados reforçam a viabilidade da DAL como uma solução escalável e econômica para a classificação de plantas daninhas em tempo real na agricultura de precisão. Trabalhos futuros estenderão essa estrutura a modelos multimodais usando imagens e metadados textuais, e este resumo representa o primeiro passo de uma investigação de doutorado mais ampla sobre DAL na agricultura de precisão.

Palavras-chave: Aprendizagem Ativa Profunda, Classificação de Ervas Daninhas, Agricultura de Precisão

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List of Abbreviations

AI Artificial Intelligence

ML Machine Learning

DL Deep Learning

AL Active Learning

DAL Deep Active Learning

CNN Convolutional Neural Networks

UAV Unmanned Aerial Vehicle

CV Computer Vision

SIS Semantic Image Segmentation

RGB Red, Green and Blue

TP True Positive

FN False Negative

FP False Positive

TN True Negative

Introduction

1.1 *Background and Motivation*

Precision agriculture utilizes advanced technologies, including GPS, IoT sensors, and remote sensing, to optimize field-level management of crops. Precision agriculture faces challenges despite its potential to increase yield and resource-use efficiency, including heterogeneous data sources, environmental variability, and high implementation costs [Li et al., 2023; Oghaz et al., 2019].

Modern precision agriculture increasingly relies on deep learning (DL) techniques to automate decision-making and reduce human labor [Osco et al., 2021; Milioto et al., 2018]. This includes convolutional neural networks (CNNs) for classification, object-detection models for weed and pest localization, encoder-decoder architectures for semantic segmentation, and density-map estimators for counting applications. Despite their effectiveness, DL models typically require large volumes of manually labeled images to achieve high performance, rendering dataset creation both time-consuming and expensive. This requirement for extensive annotation poses a significant barrier to widespread adoption in agricultural settings [Li et al., 2023; Chiu et al., 2020].

Active learning (AL) offers a promising approach by selectively querying the most informative samples from an unlabeled dataset, thereby reducing the overall annotation effort needed to develop accurate models in precision agriculture contexts [Zahidi and Cielniak, 2021; van Marrewijk et al., 2024]. Deep active learning (DAL), which integrates data sampling with model optimization, has improved performance compared to traditional acquisition active learning (AL) strategies. This method more effectively leverages model uncertainty and

representation learning, facilitating faster convergence with fewer annotations [Ge et al., 2024; Sener and Savarese, 2017].

In image classification, DAL methods have been successfully applied to tasks such as crop-weed discrimination and row detection in plantations using UAV imagery. These methods achieve accuracy levels comparable to fully supervised approaches while requiring annotations for only a small portion of the data [Osco et al., 2021; Zahidi and Cielniak, 2021]. Furthermore, DAL strategies have been successfully implemented in semantic segmentation on extensive aerial datasets and field phenotyping studies to effectively delineate crops, weeds, and anomalies. Therefore, applying DAL to semantic image segmentation problems can increase annotation efficiency and improve segmentation accuracy under real-world conditions [Malambo et al., 2019; Chiu et al., 2020].

Despite recent advancements, the targeted application of Domain Adaptation Learning (DAL) specific to the dual functions of classification and semantic segmentation in precision agriculture remains insufficiently investigated. This gap in the literature underscores the need for the current study, which aims to develop and evaluate DAL frameworks that effectively address these shortcomings [Li et al., 2023; Ge et al., 2024].

1.2 Problem Definition

In precision agriculture, classifying weed species within agricultural imagery poses a significant challenge due to the considerable variability in plant appearance caused by differing growth stages, lighting conditions, and environmental factors. Furthermore, the subtle visual differences among various weed classes underscore the need for advanced feature extraction and classification techniques to ensure reliable discrimination between similar species [Goyal et al., 2025].

The annotation process for large-scale agricultural image datasets is labor-intensive, time-consuming, and costly, often requiring significant domain expertise to accurately differentiate between visually similar classes, such as crops and weeds [Wang and Stavness, 2025; Rawat et al., 2022]. Tasks that involve semantic segmentation, where each pixel must be labeled individually, are particularly challenging, prone to errors, and resource-intensive [van Marrewijk et al., 2024; Wang and Stavness, 2025]. Furthermore, agricultural datasets frequently exhibit class imbalance, with many annotated images contributing little to model learning, which diminishes annotation efficiency and escalates costs [Wang and Stavness, 2025]. These challenges render manual annotation at scale increasingly unsustainable, often resulting in diminished

dataset quality due to mislabeling and inconsistencies [Silva et al., 2024].

The limited availability of labeled data poses a significant challenge in precision agriculture classification tasks, which are often further complicated by substantial class imbalances and noisy or ambiguous samples. These issues can adversely affect model performance [Johnson and Khoshgoftaar, 2019]. To effectively tackle these challenges, it is essential to integrate advanced deep learning architectures with specialized active learning methodologies specifically designed to handle imbalanced and noisy datasets. Such comprehensive computational strategies have demonstrated promising results in enhancing the robustness and generalization capabilities of models when applied to various real-world agricultural environments [Ren et al., 2021].

1.3 Hypothesis

In this work, we have the following hypotheses:

1. **Deep Active Learning in Agriculture Applications Improves Accuracy with Less Labeled Data:** We hypothesize that utilizing Data-Active Learning (DAL) strategies will markedly enhance the accuracy of applications in precision agriculture when compared to conventional supervised learning methods, all while necessitating a reduced volume of labeled data. This improvement stems from the DAL’s ability to intelligently identify the most informative samples for annotation, thereby directing labeling efforts toward the most challenging and ambiguous cases. In contrast, standard supervised learning often relies on random sampling, which may include a significant number of redundant or easily classified examples.
2. **Multimodal Models (Text+Images) Improve Deep Active Learning Methods** Integrating multimodal information, particularly the combination of text and images, can significantly improve the performance of Deep Active Learning (DAL) methods in image classification tasks. This improvement is anticipated due to the enhanced feature representations and contextual understanding that arise from merging visual and textual data. Such integration may lead to more effective selection of informative samples for annotation, as highlighted by recent advancements in vision-language models [Bang et al., 2024].

1.4 Objectives

The primary objective of this work is to evaluate DAL methods applied to RGB images in agricultural contexts, aiming to develop strategies that reduce the amount of labeled data required while maintaining classification accuracy. To achieve this overall goal, we have defined the following specific objectives:

1. **Dataset Creation and Annotation:** Construct datasets of high-resolution RGB images captured by UAVs for various precision agriculture applications - such as weed classification and anthill segmentation—ensuring representation across different environmental conditions, phenological stages, and spatial contexts. This process will involve field data collection and manual annotation tailored to the specific task requirements, such as classification labels or segmentation masks.
2. **Baseline Model Development and Evaluation:** Establish baseline performance by training and evaluating state-of-the-art CNN architectures on a fully labeled and randomly sampled subset of the dataset. Performance will be assessed using standard quantitative metrics, such as accuracy, precision, recall, and F1-score, to serve as a reference for comparisons with Deep Active Learning strategies.
3. **Implementation and Evaluation of Uncertainty-Based Sample Selection in DAL:** Implement and compare multiple uncertainty-driven acquisition functions within a Deep Active Learning (DAL) framework. CNN models will be iteratively trained using samples selected by each acquisition strategy, and their performance will be evaluated using the same metrics applied to the baseline, enabling a systematic comparison of sample selection efficiency.
4. **Labeling Effort Reduction and Analysis of Selected Samples:** Quantify the reduction in labeling effort achieved by the best-performing Deep Active Learning strategy relative to the baseline supervised approach. Additionally, analyze the characteristics of the selected samples through visualization, clustering, or dimensionality reduction techniques to better understand the selection behavior and identify challenging or informative patterns in the data.
5. **Development of a Multimodal Deep Active Learning Strategy:** Propose and evaluate a Deep Active Learning strategy that integrates multimodal data by combining RGB UAV imagery with textual or contextual metadata to guide the sample selection process. The approach aims

to leverage the complementary strengths of visual and textual information to enhance annotation efficiency and model performance in precision agriculture tasks.

1.5 Dissertation Text Organization

This section presents the organization of this thesis qualification proposal.

This thesis is organized into three main chapters. Chapter 1 presents the introductory sections of this qualification work, addressing the background and motivation in Section 1.1, the definition of the problem in Section 1.2, the hypotheses in Section 1.3, the general and specific objectives in Section 1.4 and the structure of the dissertation in Section 1.5.

Chapter 2 is a scientific paper developed during the implementation of the objectives of this work, where the main results of the first part of this thesis qualification work are presented. In this Chapter 2, it is possible to find the entire methodology of this scientific process used, the definition of the problem proposed in the chapter, related works, methodology, definition of the data sets used and the DAL techniques evaluated, ablation results, quantitative and qualitative results, in addition to the general discussion of the chapter and limitations found in the process. Finally, in Chapter 3 we present the conclusions of this qualification work to date, including a parallel between the objectives of this dissertation and the results obtained, future works and the execution schedule of the following activities to be developed until the completion of the final thesis.

Deep Active Learning for Weed Species Classification in Images

This chapter presents the paper submitted to **IEEE Geoscience and Remote Sensing Letters**.

Abstract

Accurate weed identification is crucial for precision agriculture; however, it presents significant challenges due to visual variations influenced by growth stage, species, environment, and image capture techniques, which typically require extensive manual annotation. To tackle the issue of annotation costs, our research investigates Deep Active Learning (DAL) for classifying weeds into multiple categories. We developed the DalMax framework, which incorporates a ResNet-50 convolutional neural network (CNN) trained on an agricultural image dataset and evaluates various active learning strategies, including MarginSampling, BALDDropout, EntropySampling, LeastConfidence, and others. Our approach commenced with 100 labeled images and iteratively enriched the dataset over 8 rounds by querying informative samples in batches of 10, 50, and 100. The performance of DAL models was evaluated both quantitatively and qualitatively based on the hyperparameters tested and the number of images collected per round. Margin sampling with a query size of 100 yielded an F1-score of 89.14%, approaching the 91.34% upper bound achieved by training on the complete dataset of 8,086 images. Confusion matrices and visual analyses demonstrated that DAL strategies consistently enhanced accuracy compared to random selection, although certain classes with subtle

distinctions remained difficult to classify. These findings underscore the potential of DAL, particularly MarginSampling, to significantly reduce annotation efforts without substantial loss in accuracy, attaining near upper-bound performance with approximately 86% fewer labeled images. Our findings emphasize the viability of DAL as a scalable and cost-effective solution for real-time weed classification, thereby promoting sustainable agricultural practices.

2.1 *Introduction*

Precision agriculture is increasingly leveraging advanced image analysis techniques to enhance crop management and promote sustainable farming practices. A critical component of this approach is the accurate identification and classification of weed species, which is essential for optimizing herbicide application and minimizing environmental impact [Zhang et al., 2023]. These automated systems enhance decision-making processes, resulting in more efficient and environmentally friendly agricultural operations. Weed infestations pose a persistent and significant threat to global agriculture, resulting in substantial losses of crop yield. Traditional weed management methods, such as manual removal and extensive use of herbicides, are labor-intensive, environmentally harmful, and economically unsustainable. In contrast, precision agriculture provides a promising alternative by facilitating targeted weed control, thereby reducing chemical use and promoting overall crop health [Hasan et al., 2023].

Accurate identification and classification of weed species are essential for the success of precision agriculture. The visual similarities between weeds and crops, variations in growth stages, and diverse environmental conditions, however, complicate this process. Conventional supervised learning models, which form the basis of many image classification tasks, require extensive labeled datasets, which are often limited in agricultural contexts due to the high costs and time associated with data annotation [Hasan et al., 2021].

Recent advancements in deep learning (DL) have revolutionized image classification tasks, achieving notable accuracy and robustness. Convolutional Neural Networks (CNNs) and other deep architectures have proven highly effective at extracting complex features from images, which is particularly advantageous for the intricate task of weed classification [Sishodia et al., 2020; Weiss et al., 2020; Adhinata et al., 2024]. Nonetheless, these models generally require large amounts of labeled data, presenting significant challenges due to the substantial costs and labor involved in manual annotation [Dou et al., 2021]. Furthermore, the training of the CNN models is computationally expensive due to the large volume of data [Peng et al., 2022].

Deep active learning (DAL) effectively solves the data annotation bottleneck by selectively querying the most informative samples for labeling [Li et al., 2024a]. Such an approach reduces labeling effort while enhancing model performance, particularly in multiclass classification tasks where inter-class variation can be subtle and complex [Li et al., 2024a]. Integrating active learning with deep learning frameworks emerges as a promising strategy to improve the efficiency and accuracy of weed classification systems in practical agricultural applications.

DL models have demonstrated significant potential in classifying plant images [Carvalho et al., 2022; Joshi et al., 2025]. Their effectiveness, nevertheless, largely depends on the availability of extensive, well-annotated datasets, which are often scarce in the agricultural sector. The resulting scarcity can hinder the model's ability to generalize across diverse conditions. Moreover, a notable research gap exists in the application of DAL strategies aimed at reducing reliance on large labeled datasets, particularly for weed classification [Adhinata et al., 2024]. While existing datasets, such as DeepWeeds [Olsen et al., 2019] and Weed25 [Wang et al., 2022], demonstrate the potential of DL in this field, they also underscore the urgent need for innovative methods that can reduce dependence on extensive labeled data.

Deep learning models that accurately classify weed species can be crucial in a global landscape where precision agriculture is becoming increasingly implemented to enhance productivity and sustainability. However, their scalability is frequently hindered by the need for large amounts of labeled data, which may not be easily accessible in many areas, especially in developing countries [Adhinata et al., 2024].

The primary aim of our research is to develop a DL model for the multiclass classification of weed plant images using DAL methods. It seeks to integrate active learning strategies to reduce the dependence on large labeled datasets and to assess the model's performance across various environmental conditions and weed species. In essence, the primary objective of this research is to propose a computational approach that combines DL and DAL techniques for the efficient and accurate classification of multiple weed species. This approach aims to reduce reliance on extensive labeled datasets, thereby contributing to the broader accessibility and scalability of precision agriculture technologies worldwide.

2.1.1 Chapter Organization

The chapter is organized as follows. The section above clearly presents the background and motivation, defining the research problem and outlining the primary contributions. This establishes a comprehensive framework for the

study, following practical guidelines for scientific communication in computer science [Shaw, 2003; Theisen et al., 2017].

Section 2.2 provides a review of the related literature on DL and active learning methodologies, with a focus on their applications to precision agriculture and weed classification tasks. This section synthesizes insights from recent research, identifying current advances and explicitly highlighting knowledge gaps targeted by our study [Gastel and Day, 2019].

The subsequent sections outline the methodology, experimental design, and results, offering a comprehensive discussion of the findings. The paper concludes by summarizing key results and suggesting directions for future research. This structured approach ensures logical flow, clarity, and coherence, consistent with established best practices in scientific writing [Gastel and Day, 2019].

2.2 *Related Work*

2.2.1 *Deep Learning in Image Classification*

Deep learning has fundamentally revolutionized image classification tasks, driving substantial advancements across various domains, including medical imaging, autonomous vehicles, and remote sensing [LeCun et al., 2015]. CNNs, in particular, have played a pivotal role, consistently establishing new benchmarks for accuracy and robustness in visual recognition tasks [Alom et al., 2019]. These models have significantly benefited from architectural innovations and the availability of large-scale datasets, resulting in enhanced capabilities for feature extraction, generalization, and reliable performance across diverse and challenging scenarios [Khan et al., 2020].

Recent advancements in convolutional neural network (CNN) architectures have significantly enhanced the capabilities of feature extraction and representation. Innovations such as residual connections, attention mechanisms, and depthwise separable convolutions have contributed to the development of more efficient and effective models [He et al., 2016; Dosovitskiy et al., 2020; Wang et al., 2025]. For instance, the integration of residual connections in ResNet architectures has addressed the vanishing gradient problem, enabling the training of deeper networks and improving performance in high-dimensional image datasets [He et al., 2016]. Attention mechanisms, likewise, as implemented in models such as SENet, have enabled networks to focus on the most informative features, thereby enhancing accuracy [Hu et al., 2018]. Moreover, integrating depthwise separable convolutions in architectures such as MobileNet has reduced computational complexity while maintaining robust performance, making them suitable for deployment on resource-constrained

devices [Howard et al., 2017].

Recent research has explored the integration of novel data augmentation and regularization techniques to address challenges related to data scarcity and class imbalance [Shorten and Khoshgoftaar, 2019]. Compelling evidence demonstrates that advanced augmentation strategies not only mitigate overfitting but also enhance the generalization capabilities of DL models across diverse application scenarios [Shorten and Khoshgoftaar, 2019].

In addition to architectural and data-centric enhancements, the development of efficient training algorithms has become a focal point in addressing the computational demands of deep neural networks. Notably, the advent of models such as MobileNet has demonstrated that it is possible to achieve high accuracy with significantly fewer parameters, making it suitable for deployment in resource-constrained environments [Howard et al., 2017]. Equally important, the introduction of EfficientNet has demonstrated the efficacy of compound scaling, which involves uniformly scaling network depth, width, and resolution to achieve superior performance with reduced computational resources [Tan and Le, 2019]. These advancements offer promising solutions for deploying high-performance image classification models in resource-constrained environments. Some works have introduced a novel training framework that leverages model improvement and transfer learning, offering a promising solution for deploying high-performance image classification models in resource-constrained environments [Yousif, 2024; Liu et al., 2023].

The classification task in this study employs the ResNet50 convolutional neural network, a widely recognized architecture known for its excellent performance and ability to mitigate the vanishing gradient problem through residual connections [He et al., 2016]. ResNet50 has demonstrated strong generalization capabilities across various image classification tasks, making it particularly suitable for evaluating the effectiveness of samples selected by deep active learning techniques [Alom et al., 2019]. ResNet50 provides a reliable classifier backbone for evaluating weed image selection, enabling accurate assessment of active learning strategies' impact on model accuracy and efficiency.

Beyond standard deep convolutional architectures, recent advancements in feature extraction have demonstrated the high efficiency of randomized neural networks for capturing complex spatio-spectral patterns. Methods such as the Spatio-Spectral Randomized Autoencoder (SSRAE) [Fares and Ribas, 2024] and the Volumetric Color-Texture Representation (VCTex) [Fares and Ribas, 2025] have emerged as highly discriminative, single-parameter texture representations. Leveraging randomized weights and closed-form solutions rather than expensive iterative training allows these methods to extract ro-

bust spatial and cross-channel color information with significantly reduced computational costs. This makes them particularly well-suited for encoding high-resolution images and providing a rich, descriptive feature space for subsequent classification tasks.

2.2.2 *Active Learning and Deep Active Learning Techniques*

Deep active learning (DAL) has emerged as a crucial strategy to alleviate the high annotation costs associated with deep neural networks, significantly enhancing model efficiency in image classification tasks [Li et al., 2024a]. Foundational uncertainty-based acquisition functions prioritize samples where the model is least confident. Specifically, Least Confidence selects instances with the lowest maximum prediction probability [Lewis and Gale, 1994], while Margin Sampling measures the gap between the top two predicted classes to identify highly ambiguous instances [Scheffer et al., 2001]. Beyond these heuristics, EntropySampling evaluates the information content across the entire label distribution, prioritizing instances with maximum overall predictive uncertainty [Settles, 2009]. Although considered traditional, these deterministic metrics remain highly effective baselines in modern DAL applications, particularly in precision agriculture, where distinguishing morphologically similar weed species poses a significant classification challenge [Ren et al., 2021].

To adapt these foundational concepts for deep neural networks and mitigate the overconfidence typically observed in standard deterministic models, recent advancements integrate Monte Carlo dropout to approximate Bayesian inference [Gal and Ghahramani, 2015]. Maintaining dropout layers active during inference enables models to perform multiple stochastic forward passes for each unlabelled sample. The resulting probabilities are averaged to form a highly calibrated predictive distribution. Applying Monte Carlo dropout yields robust stochastic variants, namely LeastConfidenceDropout, MarginSamplingDropout, and EntropySamplingDropout, which provide a more reliable measure of epistemic uncertainty, thereby facilitating highly accurate sample selection for complex classification tasks [Kendall and Gal, 2017].

Complementary to uncertainty estimation, cluster-based selection techniques prioritize training set diversity by identifying representative samples across the data distribution. Approaches such as KCenterGreedy and KMeansSampling, foundational to the core-set methodology [Sener and Savarese, 2017], operate by minimizing the distance between the selected subset and the remaining unannotated pool. By exploring the geometric properties of the decision space, adversarial strategies can also estimate a sample’s distance to the decision boundary by leveraging the model’s sensitivity to crafted perturbations [Ducoffe and Precioso, 2018]. Techniques like AdversarialBIM apply the

Basic Iterative Method to incrementally inject gradient-signed noise, whereas AdversarialDeepFool iteratively projects the sample onto the closest linearized hyperplane to compute the minimal required perturbation [Moosavi-Dezfooli et al., 2016]. As a result, both methods effectively identify highly informative instances critically positioned near decision boundaries.

While the integration of uncertainty, diversity, and geometric heuristics has demonstrated profound potential in reducing manual annotation efforts [Joshi et al., 2012; Ranganathan et al., 2017; Li et al., 2024b], addressing the long-tail distribution of concepts in large uncurated datasets remains a critical challenge. Recently, advanced hierarchical clustering approaches have been proposed to approximate a uniform distribution over the data support, ensuring a highly balanced sample selection [Vo et al., 2024]. Concurrently, spatio-spectral feature extractors like SSRAE and VCTex have proven highly efficient at capturing complex image patterns. Despite this, their integration within a deep active learning framework remains unexplored. To bridge this gap, this study introduces two novel acquisition functions: SSRAEKmeansHCSampling and VCTexKmeansHCSampling. The application of a hierarchical K-means sampling strategy directly to the comprehensive embeddings generated by SSRAE and VCTex, allows these proposed methods to select the most informative, diverse, and conceptually balanced subsets from the unlabeled pool.

To systematically evaluate these paradigms, our experimental framework implements fourteen distinct acquisition functions. RandomSampling serves as the standard baseline. Deterministic uncertainty is assessed through Least-Confidence, MarginSampling, and EntropySampling, while epistemic uncertainty is captured via their stochastic counterparts and BALDDropout (Bayesian Active Learning by Disagreement) [Gal et al., 2017]. Diversity-driven selection is represented by KCenterGreedy and KMeansSampling, and geometric exploration utilizes AdversarialBIM and AdversarialDeepFool. Finally, extending the concepts of diversity and representation learning, our novel approaches, SSRAEKmeansHCSampling and VCTexKmeansHCSampling, uniquely combine advanced spatio-spectral feature extraction with hierarchical clustering to establish a new standard for balanced and informative sample selection [Fares and Ribas, 2024, 2025; Vo et al., 2024].

2.2.3 *Applications in Agriculture and Weed Classification*

Recent advances in DL have revolutionized the field of precision agriculture, particularly in automated weed detection and classification. Convolutional neural network structures can effectively distinguish weed species from crop images, demonstrating high accuracy even under variable field conditions

[Abdulsalam and Aouf, 2020; Pai et al., 2024]. Such automation minimizes manual work and also improves decision-making in crop management.

The integration of unmanned aerial vehicle (UAV) imaging with DL techniques has significantly advanced agricultural applications. For instance, a study utilized UAV-acquired RGB images to compare various DL segmentation models for weed detection, identifying that the UNet [Ronneberger et al., 2015] model with EfficientNetB0 [Tan and Le, 2019] as a backbone achieved high precision and recall rates, thereby enhancing the efficiency of weed management in agricultural settings [Pai et al., 2024]. Parallel studies have demonstrated that combining UAV technology with CNNs enables accurate and efficient weed detection, highlighting the potential for the scalable and practical deployment of DL methods in real-world farming scenarios [Zhang et al., 2024a].

Visual characteristics, such as color, texture, and shape, inherent in RGB images, are crucial for distinguishing between weeds and crops, providing significant advantages in the classification of invasive plants. The joint analysis of color and texture in RGB images can achieve considerable accuracy in weed identification [Mekhalfa and Yacef, 2021]. Integrating color and texture proves particularly useful in scenarios where the image is simply a traditional RGB image captured via a UAV.

The active learning strategy in the agricultural sector holds promise for optimizing resources and reducing costs associated with data annotation [van Marrewijk et al., 2024]. In contrast to the practice of exhaustively labeling the entire dataset, active learning focuses on intelligently selecting the most informative samples for annotation [Settles, 2009]. Focusing on targeted selection not only minimizes the manual effort and costs associated with annotating large volumes of data but also ensures that the model learns more efficiently by focusing on the information most relevant to the task at hand, such as detecting crop diseases or identifying invasive plants [van Marrewijk et al., 2024]. In this way, active learning can facilitate more effective resource management and informed decision-making in the agricultural sector.

2.3 *Materials and Methods*

2.3.1 *Study Area and Data Acquisition*

The images were captured using a UAV, which enabled the acquisition of high-resolution agricultural scenes and the creation of an orthomosaic. Different regions of Brazil were surveyed, covering 282 farms located in the states of São Paulo and Mato Grosso do Sul, Brazil. The exact locations of these regions have been omitted due to contractual privacy agreements. The or-

thomosaics were partitioned into non-overlapping patches of 128×128 pixels. Subsequently, specialists labeled only those patches containing weeds, classifying each into one of the C weed classes to ensure accurate annotations. All images were stored in RGB format. A comprehensive dataset of 10,192 images was thus created and organized into two primary directories: an Unlabeled Pool set containing 8,086 images and a fixed Labeled Test set comprising 2,106 images. Five distinct weed species classes were identified: Gramineae spp., *Ricinus communis*, *Brachiaria* spp., *Panicum maximum*, and broad-leaved species. Figure 2.1 illustrates representative examples of these weeds under realistic environmental conditions.

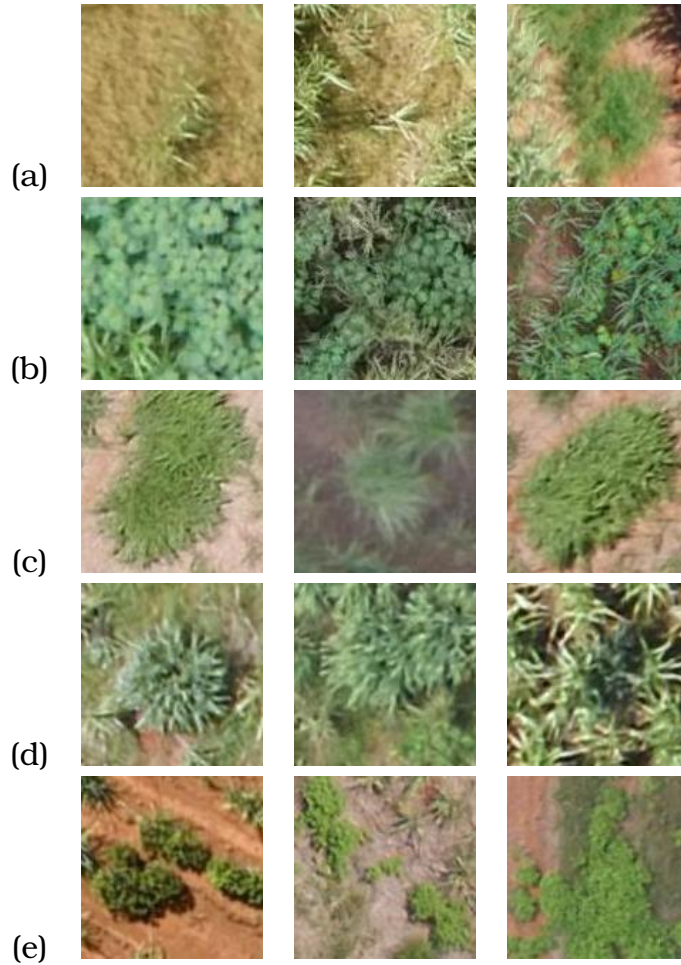


Figure 2.1: Representative sample images of each class in the dataset under realistic environmental conditions, namely: (a) Gramineae spp., (b) *Ricinus communis*, (c) *Brachiaria* spp, (d) *Panicum maximum* and (e) Broadleaf species. Source: Carvalho, 2024.

In the context of DAL, the Pool set is intended to serve as a reservoir of unlabeled images from which the model actively selects the most informative samples for annotation during the experiment. To simulate the active learning process without incurring excessive manual labor, all images in the Pool set were pre-annotated. Pre-annotating the pool enabled us to efficiently mimic

the gradual annotation process that typically occurs as images are selected for training.

2.3.2 Image Annotation and Preprocessing

The annotation process was meticulously performed by a domain expert who manually reviewed each image and correctly assigned it to its respective class folder based on the weed species. Through meticulous manual labeling, we ensured that the dataset maintained high labeling accuracy, which is critical for training robust deep active learning models. The clear segregation of images into class-specific directories has streamlined subsequent data handling and analysis. Accordingly, during the experiments, although the Pool set was treated as unlabeled for active selection, the availability of pre-annotated images enabled rapid and accurate updates to the training base, facilitating an effective evaluation of the DAL methods. The distribution of images by class for both the Pool (annotated post-selection) and Test sets is detailed in Table 2.1.

Dataset class	Size Pool set	Size Test set
Gramineae spp.	1,198	970
Ricinus communis	3,370	456
Brachiaria spp.	774	335
Panicum maximum	1,036	190
Broadleaf species	1,708	155
Total	8,086	2,106

Table 2.1: Distribution of classes in the annotated data set.

All images were normalized using the standard ImageNet weights [Deng et al., 2009]. Specifically, each image was scaled and normalized based on the mean and standard deviation computed from the ImageNet dataset [Deng et al., 2009]. Standardizing the data through this normalization step is a widely adopted practice in DL pipelines [Ioffe and Szegedy, 2015] that helps reduce the effects of varying illumination and color discrepancies across the dataset.

2.3.3 Proposed Methodology

The methodological framework of this study involved acquiring high-resolution imagery data using an unmanned aerial vehicle (UAV), followed by the specialized manual annotation of images to create a high-quality initial dataset. This dataset was integrated into the DalMax platform [Carvalho, 2024], an open-source deep active learning (DAL) framework implemented in Python 3¹.

¹<https://www.python.org/>

DalMax enables flexible experimentation with a wide range of active learning strategies, offering structured control over parameters such as iteration count, query size, and initial labeled set size through configuration files [Pezoa et al., 2016].

The experimental protocol was divided into two phases. In the first phase, different DAL methods were evaluated using varied parameter settings. In the second phase, an image classification task was conducted to validate the effectiveness of the DAL strategies in selecting the most informative samples. For all experiments, the ResNet-50 architecture [He et al., 2016], pre-trained on ImageNet and fine-tuned for the weed classification task, was employed as the backbone model. Minor architectural modifications were made to efficiently accommodate the iterative learning process, allowing the network to integrate newly labeled samples over successive rounds.

Each experimental execution consisted of eight active learning rounds. Initially, 100 samples were randomly selected and annotated. In each subsequent round, a new batch of samples was queried and annotated, with query sizes of 100, 50, or 10 samples, depending on the experiment. At each round, the ResNet-50 model was retrained for ten epochs using the updated dataset and evaluated on a reserved test set. Iterative labeling allowed for a dynamic assessment of model performance and sample informativeness.

To ensure robustness, three distinct experiments were conducted corresponding to the three query sizes, and each configuration was repeated three times with different random seeds. A total of 126 experimental runs were performed (14 DAL methods $\times$ 3 seeds $\times$ 3 query sizes). The primary performance metric was the F1-score, complemented by confusion matrix analyses to gain a deeper understanding of classification outcomes.

As the number of annotated samples increased with each round, the total varied according to the query size. For instance, with 100 samples per round, the final set comprised 900 annotated images. With 50 and 10 samples per round, the final totals were 500 and 180 images, respectively. In all cases, the annotated subset remained significantly smaller than the full dataset of 8086 images. For baseline comparisons, random sampling was consistently included in all experiments.

Additionally, to establish an upper performance bound, a full training session was conducted using the entire pool of images. This complete training run provided a reference for maximum achievable accuracy, facilitating the evaluation of the performance gap between random selection and DAL strategies.

The DAL methods tested included RandomSampling, LeastConfidence, MarginSampling, EntropySampling, LeastConfidenceDropout, MarginSampling-

Dropout, EntropySamplingDropout, KMeansSampling, KCenterGreedy, BALD-Dropout, AdversarialBIM, AdversarialDeepFool, SSRAEKmeansHCSampling, and VCTexKmeansHCSampling. The DalMax platform’s flexibility enabled systematic comparisons across all these methods under standardized experimental conditions.

An overview of the complete methodological workflow, from UAV-based data acquisition to model evaluation, is depicted in Figure 2.2.

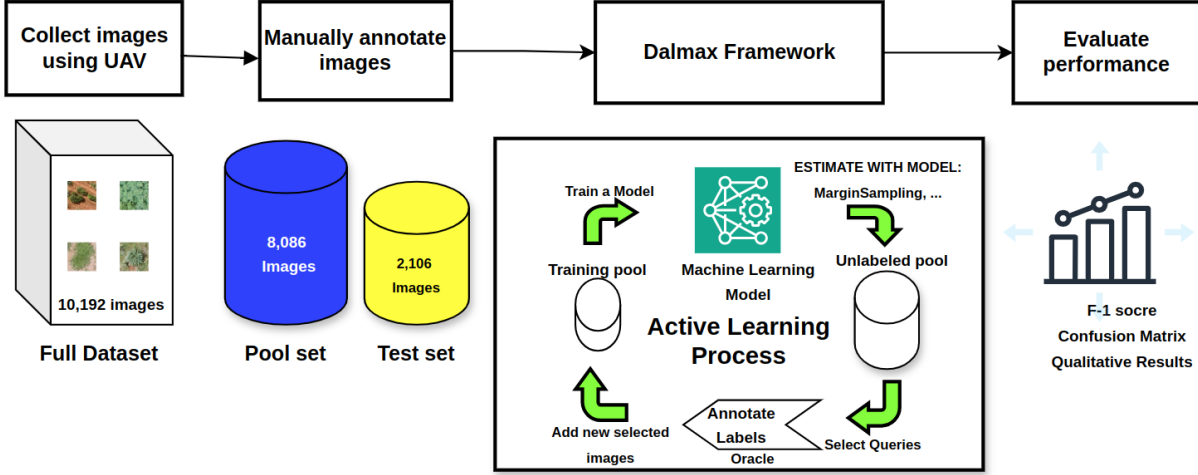


Figure 2.2: Overview of the methodological workflow employed in this study, including UAV-based data acquisition, preprocessing steps, manual annotation, integration into the DalMax framework, active learning model execution, and evaluation procedures.

2.3.4 Proposed Spatio-Spectral Hierarchical Sampling

To address the limitations of standard active learning acquisition functions, particularly their susceptibility to long-tail distributions and redundant sampling, we propose two novel query strategies: SSRAEKmeansHCSampling and VCTexKmeansHCSampling. These methods operate through a highly efficient two-stage pipeline consisting of spatio-spectral feature extraction followed by balanced hierarchical clustering.

In the first stage, the unlabelled image pool undergoes feature extraction using specialized randomized neural networks. For the SSRAEKmeansHCSampling strategy, the SSRAE is employed. This architecture independently encodes spatial arrangements within color channels and spectral relationships between adjacent channels, subsequently concatenating the flattened weights of the decoder to form a comprehensive representation [Fares and Ribas, 2024]. Alternatively, the VCTexKmeansHCSampling strategy extracts features using the VCTex. This method models complex spatio-spectral patterns simultaneously by sliding $3 \times 3 \times 3$ volumetric color cubes over the image

and encoding them via a randomized autoencoder [Fares and Ribas, 2025]. Both techniques leverage closed-form mathematical solutions rather than iterative backpropagation, yielding highly discriminative, low-dimensional feature vectors with minimal computational overhead.

Figure 2.3 illustrates the complete workflow of this initial stage, detailing how raw UAV images are independently processed by the proposed SSRAE and VCTex modules to generate highly discriminative, low-dimensional feature vectors for the subsequent active learning steps.

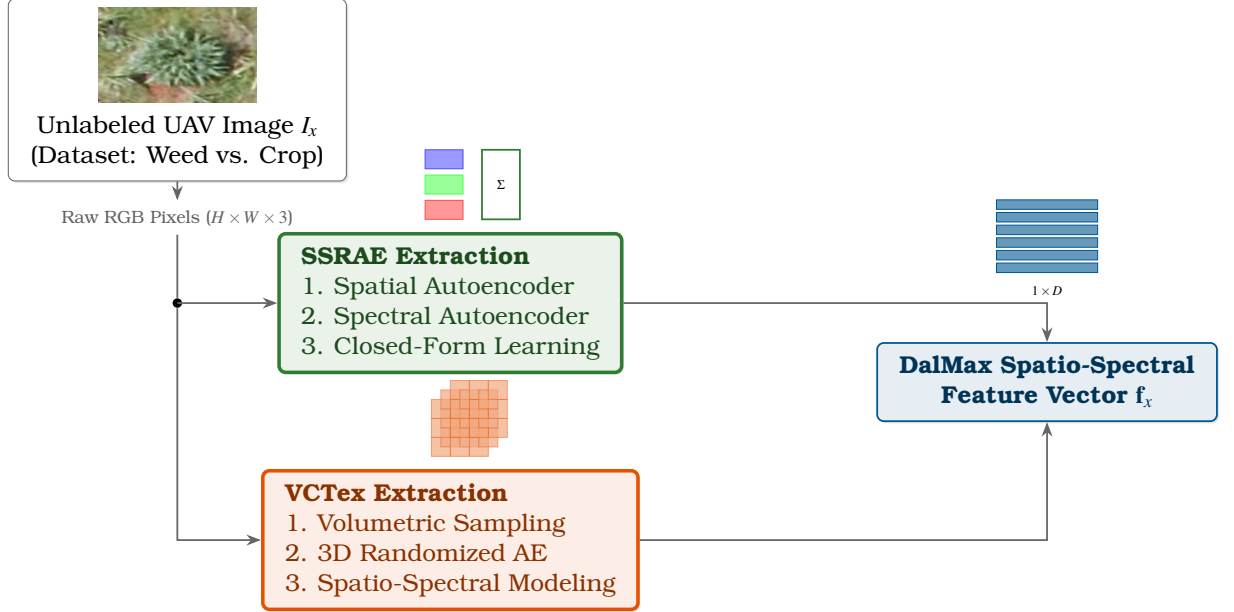


Figure 2.3: Workflow for Spatio-Spectral Feature Extraction: Comparison of the proposed SSRAE and VCTex randomized neural networks. The raw pixels from the UAV image are independently processed to generate highly discriminative, low-dimensional feature vectors that encode unique spatial and color-texture patterns.

In the second stage, a hierarchical K-means clustering algorithm is applied to the extracted feature spaces [Vo et al., 2024]. Unlike standard K-means, which tends to over-segment dense, dominant classes in imbalanced datasets, the hierarchical approach recursively clusters the data points to approximate a uniform distribution over the feature manifold. Applying multiple levels of clustering with strategic resampling helps the algorithm build a tree structure where inner nodes represent broad visual concepts and leaf nodes represent sub-concepts.

The subsequent clustering and sampling phase is detailed in Figure 2.4, which demonstrates how the Hierarchical K-means algorithm recursively partitions the long-tail distributed feature pool to extract a conceptually balanced and diverse query batch for active learning. The acquisition function samples instances uniformly across hierarchical clusters. Integrating the rich, cross-channel representations of SSRAE and VCTex with this uniform sam-

pling strategy ensures that the selected batches in the proposed methods are not only structurally diverse but also conceptually balanced, effectively mitigating biases toward majority weed classes during the active learning iterations.

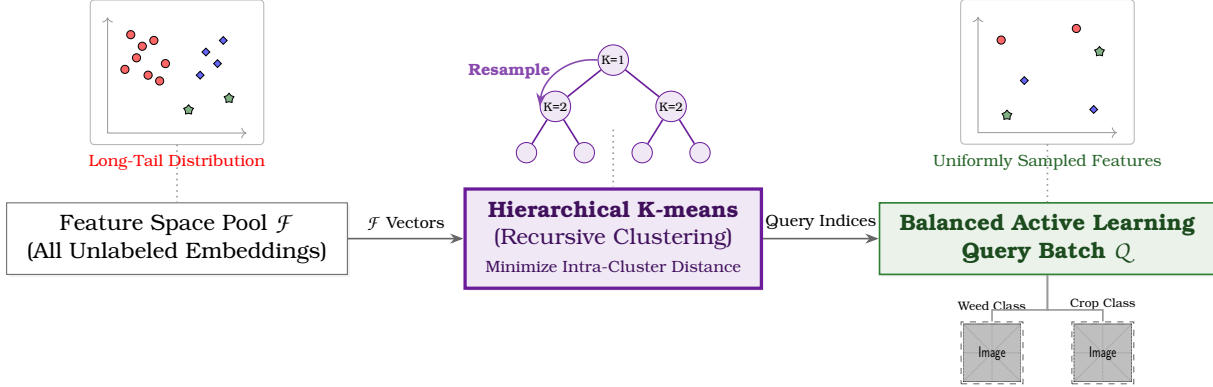


Figure 2.4: Workflow of the Proposed Spatio-Spectral Hierarchical Sampling: From an imbalanced feature pool generated by the extraction models, the Hierarchical K-means algorithm recursively clusters and resamples the embeddings to approximate a uniform concept distribution.

Figure 2.5 details the algorithm operational flow of the proposed active learning acquisition strategies. The algorithm abstracts the underlying deep learning operations into two primary phases. In the first phase, the framework iteratively processes the unlabeled image pool, extracting highly discriminative low-dimensional feature vectors utilizing either the SSRAE or VCTex method, depending on the specified configuration. In the second phase, these extracted features are organized into a conceptual tree structure via the Hierarchical K-means algorithm. Finally, a uniform sampling function retrieves a balanced batch of queries from the resulting clusters, effectively completing the active learning acquisition cycle.

2.3.5 Performance Evaluation

Model performance was evaluated using metrics derived from the confusion matrix: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) [Sokolova et al., 2009]. For this multi-class classification task, these metrics were calculated per class and then aggregated using a weighted average to account for class imbalance [Hossin and Sulaiman, 2015].

Precision, Recall and the F1-score were calculated as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.2)$$

```

Require: Unlabeled image pool  $U$ , Query size  $N$ , Extraction strategy
            $Method \in \{SSRAE, VCTex\}$ 
Ensure: Selected query batch  $Q$ 
1:  $\mathcal{F} \leftarrow \emptyset$  ▷ Initialize the feature space pool
2: for each image  $I \in U$  do
3:   if  $Method == SSRAE$  then
4:      $f_I \leftarrow \text{Extract\_SSRAE}(I)$  ▷ Encode using spatial and spectral AEs
5:   else if  $Method == VCTex$  then
6:      $f_I \leftarrow \text{Extract\_VCTex}(I)$  ▷ Encode using volumetric 3D color cubes
7:   end if
8:    $\mathcal{F} \leftarrow \mathcal{F} \cup \{f_I\}$  ▷ Append extracted vector to the feature pool
9: end for
10:  $C \leftarrow \text{Hierarchical\_KMeans}(\mathcal{F})$  ▷ Build recursive cluster tree over the
    data support
11:  $Q \leftarrow \text{Uniform\_Sample}(C, N)$  ▷ Sample  $N$  instances evenly across leaf
    clusters
12: return  $Q$ 

```

Figure 2.5: Algorithm for the Proposed Spatio-Spectral Hierarchical Sampling. The process abstracts the feature extraction via SSRAE or VCTex, followed by recursive clustering to sample a balanced conceptual query batch.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.3)$$

Due to the significant class imbalance in our dataset, the F1-score was prioritized over overall accuracy. Accuracy can be misleading in imbalanced scenarios, as it may be dominated by the performance of majority classes [Brodersen et al., 2010]. The F1-score, which considers both precision and recall, offers a more robust evaluation of model performance across all classes. The F1-score, the harmonic mean of precision and recall, provides a balanced measure of performance, particularly valuable in imbalanced datasets [Chinchor, 1992; Powers, 2011]. Confusion matrices were also generated to analyze the patterns of misclassification.

2.3.6 Hardware Configuration and Execution Details

All experiments were performed on a high-performance workstation running Ubuntu 22.04.5 LTS, equipped with 128 GiB of RAM, an AMD Ryzen 9 5950x processor (16 cores $\times$ 32 threads), and two NVIDIA GeForce RTX 3080 GPUs (10 GB each). The experiments were executed using CUDA 12.4² with NVIDIA driver version 550.120 and implemented in PyTorch [Paszke et al.,

²<https://developer.nvidia.com/cuda-toolkit>

2019]³, ensuring efficient model training and inference. Table 2.2 summarizes the hardware configuration used for all experiments.

Parameter	Specification
Operating System	Ubuntu 22.04.5 LTS
RAM	128.0 GiB
Processor	AMD® Ryzen 9 5950x (16-core × 32)
GPU	2 × NVIDIA GeForce RTX 3080 (10 GB each)
CUDA Version	12.4
Driver Version	550.120

Table 2.2: Detailed specifications of the high-performance computing environment used for training, evaluation, and testing of the DAL methods.

2.4 Experimental Results

2.4.1 Ablation Studies

Our ablation studies began by determining the top reference performance for the classification task. Training the model using the full dataset resulted in an F1-score of 0.9134 (± 0.0084), calculated from the average of three runs with different random seeds. In this section, the ablation analysis focused on the individual impact of the number of images collected per round (n_{query}), evaluating the sizes of 10, 50, and 100 images.

Figure 2.6 illustrates the F1-score evolution across 8 active learning rounds when queried with a highly restricted budget of only 10 samples per round. In this constrained scenario, learning increments are generally marginal, reflecting the inherent difficulty of extracting sufficient representative information from very small query batches. By the final round, the standard Random-Sampling baseline achieved an F1-score of 0.5746 (± 0.0471). Demonstrating the challenge of this extreme low-data regime, only three out of the thirteen evaluated active learning strategies managed to outperform this random baseline: BALDDropout (0.6293 ± 0.0429) and the two proposed hierarchical methods, SSRAEKmeansHCSampling (0.6713 ± 0.0428) and VCTexKmeansHCSampling (0.6576 ± 0.0343). Even though these strategies successfully improved sample efficiency, all methods remained considerably below the theoretical upper bound performance of 0.9134 (± 0.0084) achieved by training on the complete dataset, highlighting the severe information bottleneck imposed by the minimal sample size.

Despite these extreme budget limitations, the proposed spatio-spectral hierarchical approaches clearly established themselves as the superior strate-

³<https://pytorch.org/>

gies across the entire experiment. SSRAEKmeansHCSampling achieved the highest overall performance with a final F1-score of $0.6713 (\pm 0.0428)$, closely followed by VCTexKmeansHCSampling at $0.6576 (\pm 0.0343)$. As depicted in the learning curves, while traditional uncertainty heuristics and geometry-based methods struggled with high variance and instability, the proposed methods demonstrated a much more consistent and robust upward trajectory, pulling away from the other techniques in the later rounds. This significant performance gap underscores the efficacy of combining advanced randomized neural network feature extraction with hierarchical balanced sampling. Ensuring that even a tiny budget of 10 samples is structurally diverse and conceptually balanced enables the proposed pipelines to effectively mitigate redundancy and maximize the informational value of every single queried image.

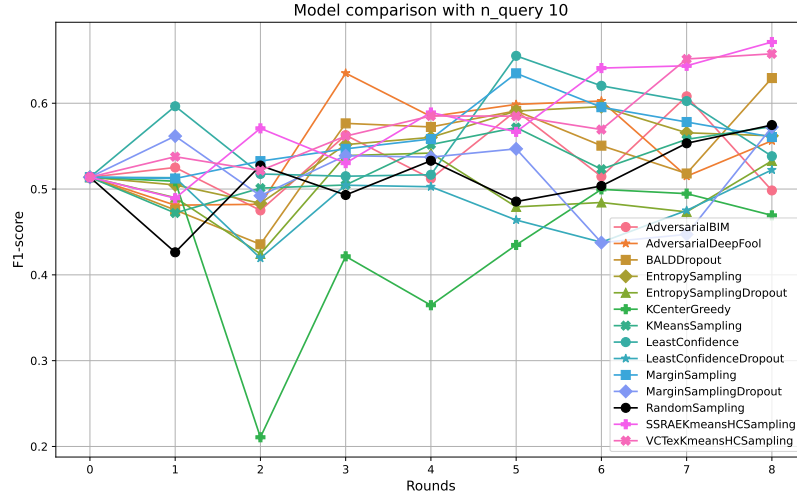


Figure 2.6: F1-score progression over 8 rounds for a query size of 10 samples per round.

Figure 2.7 presents the results for experiments with a moderate query budget of 50 samples per round. In this scenario, the performance gap between active learning strategies and the baseline becomes highly pronounced. As depicted in the learning curves, the RandomSampling approach struggled with significant instability throughout the training cycles, concluding the 8th round with a relatively low and highly variable F1-score of $0.5387 (\pm 0.2573)$. The vast majority of the evaluated DAL methods subsequently successfully outperformed the random baseline, demonstrating that a moderate increase in query size effectively accelerates the convergence of informed sampling strategies. Among the traditional techniques, MarginSampling emerged as the strongest conventional baseline, achieving a notable F1-score of $0.8023 (\pm 0.0272)$.

Surpassing all traditional and geometric heuristics, the proposed spatio-

spectral hierarchical approaches once again achieved the highest overall performance. SSRAEKmeansHCSampling reached an impressive peak F1-score of 0.8301 (± 0.0298), while VCTexKmeansHCSampling followed closely with 0.8134 (± 0.0255). Notably, the robust F1-score of 0.8301 attained by the SSRAEKmeansHCSampling strategy represents a massive gain in annotation efficiency. While the theoretical upper-bound performance achieved by training on the full dataset of 8,086 images is 0.9134 (± 0.0084), the proposed hierarchical strategy achieves competitive, near-optimal performance using only 500 annotated images (the 100 initial samples plus 50 images queried across 8 rounds). This highlights the profound capability of the proposed methods to construct a highly representative and balanced training set with substantially reduced manual annotation effort.

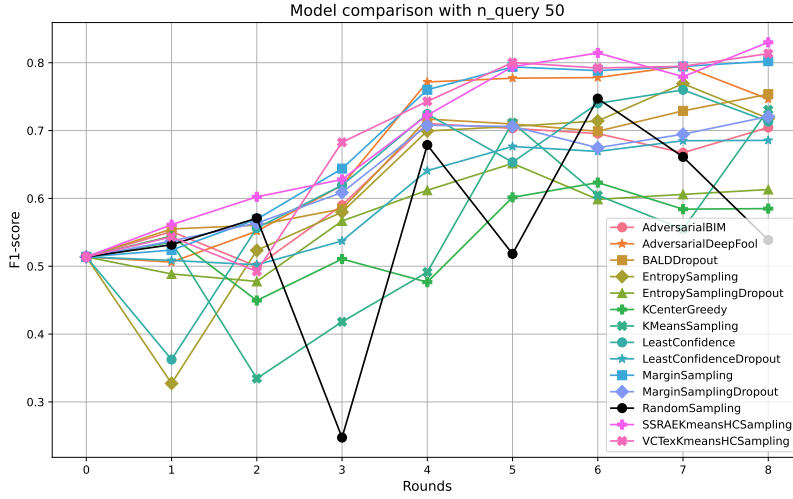


Figure 2.7: F1-score progression over 8 rounds for a query size of 50 samples per round.

The results with 100 queries per round, as shown in Figure 2.8, indicate that the advantages of active learning become highly consistent and substantial with larger query budgets. Given the increased volume of information acquired per iteration, several traditional and adversarial methods successfully surpassed the RandomSampling baseline, which reached a final F1-score of 0.8218 (± 0.0119). Among these established techniques, EntropySampling demonstrated the strongest performance, establishing a rigorous heuristic baseline by achieving an F1-score of 0.8652 (± 0.0243). These findings demonstrate that increasing the number of queried samples per round significantly stabilizes and enhances the overall effectiveness of the active learning process.

Building upon this trend of improved convergence, the proposed hierarchical strategies once again delivered the highest overall performances across the entire experimental framework. SSRAEKmeansHCSampling achieved the

peak F1-score of $0.8766 (\pm 0.0208)$, while VCTexKmeansHCSampling followed closely with $0.8712 (\pm 0.0146)$. These findings vividly illustrate the efficiency of combining spatio-spectral features with balanced clustering at scale. Notably, the absolute peak F1-score of 0.8766 achieved by the SSRAE-based method approached the theoretical upper-bound performance of $0.9134 (\pm 0.0084)$, which requires training on the complete dataset of 8,086 images-while utilizing a total of only 900 annotated images (the 100 initial samples plus 100 images queried across 8 rounds). This underscores the remarkable capacity of the proposed active learning pipelines to achieve near-optimal classification performance utilizing merely $\sim 11.13\%$ of the total dataset, thereby drastically minimizing the required manual annotation effort.

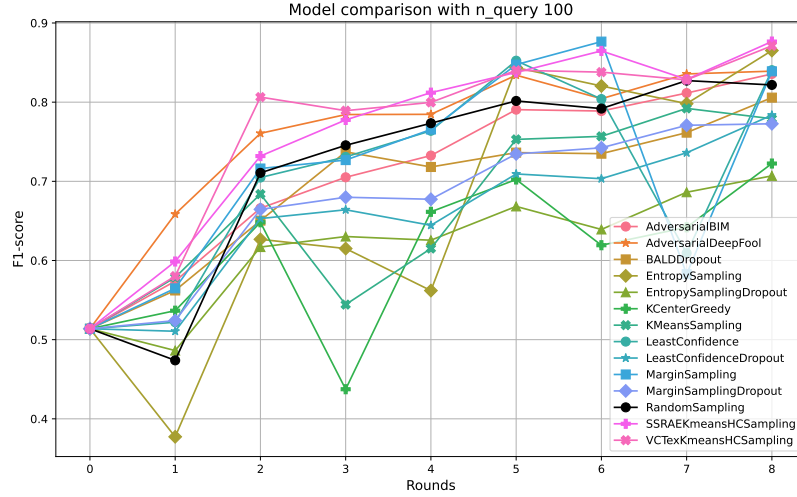


Figure 2.8: F1-score progression over 8 rounds for a query size of 100 samples per round.

Overall, the ablation studies reveal that while standard DAL heuristics may experience initial instability compared to RandomSampling, informed acquisition ultimately yields substantial gains in classification accuracy. This initial volatility in uncertainty-based sampling is a well-documented phenomenon; as noted by Ren et al. [2021], models in the earliest active learning rounds often lack a adequately calibrated feature space to produce reliable uncertainty estimates, temporarily making random selection more stable. Most importantly, this comprehensive evaluation validates the robustness and consistent superiority of the proposed hierarchical spatio-spectral methods across all evaluated query budgets. Effectively capturing complex visual patterns and mitigating long-tail class imbalances through recursive clustering allowed SSRAEKmeansHCSampling and VCTexKmeansHCSampling to consistently outperform both the random baseline and traditional DAL techniques. These results align closely with recent findings in the literature, which demonstrate

that traditional uncertainty heuristics frequently suffer from severe sampling redundancy in long-tailed distributions, repeatedly querying visually similar majority-class instances while ignoring diverse minority concepts [Geifman and El-Yaniv, 2017; Ren et al., 2021; Singh et al., 2022; Hayes et al., 2022; Lesci and Vlachos, 2024; van Marrewijk et al., 2024]. These findings conclusively demonstrate that adopting advanced, conceptually balanced active learning pipelines not only solves the redundancy problem in multi-class datasets, but also maximizes overall annotation efficiency in complex agricultural scenarios.

2.4.2 Quantitative Results

Initially, it is essential to establish that training the ResNet-50 classifier on the entire pool of 8,086 images yielded an upper-bound F1-score of 0.9134 (± 0.0084) across three different random seeds. This represents the maximum achievable performance on this dataset and serves as a fundamental benchmark for our experiments. The subsequent quantitative results evaluate how closely the varied active learning strategies can approach this upper bound while utilizing significantly fewer manually labeled samples. The best-performing method and random sampling are bolded in Tables 2.3, 2.4, and 2.5.

For the highly constrained scenario with $n_{query} = 10$, Table 2.3 reports the comprehensive classification metrics accuracy, precision, recall and F1-score, achieved after 8 active learning rounds. Focusing on the primary metric of interest, the F1-score, the RandomSampling baseline concluded the process with an average of 0.5746 (± 0.0471). Given the minimal sample size queried per round, several traditional heuristics struggled to consistently outperform this random selection. In contrast, the proposed hierarchical methods demonstrated clear and absolute superiority. SSRAEKmeansHCSampling achieved the highest overall performance with an F1-score of 0.6713 (± 0.0428), followed closely by VCTexKmeansHCSampling at 0.6576 (± 0.0343). Among the established DAL techniques, BALDDropout was the only traditional method to notably surpass the baseline, recording an F1-score of 0.6293 (± 0.0429). These quantitative results validate that, even under severely restricted annotation budgets, the proposed spatio-spectral hierarchical sampling successfully selects a significantly more informative and balanced query batch compared to standard uncertainty-based methods and random sampling.

Expanding the annotation budget to $n_{query} = 50$, Table 2.4 details the classification metrics across all evaluated methods. In this moderately scaled scenario, the performance gap between active learning strategies and random selection widened considerably. The RandomSampling baseline exhibited signif-

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.4789 (± 0.0460)	0.7362 (± 0.0534)	0.4789 (± 0.0460)	0.4984 (± 0.0518)
AdversarialDeepFool	0.5442 (± 0.0416)	0.7231 (± 0.0442)	0.5442 (± 0.0416)	0.5563 (± 0.0362)
BALDDropout	0.6223 (± 0.0466)	0.7152 (± 0.0174)	0.6223 (± 0.0466)	0.6293 (± 0.0429)
EntropySampling	0.5514 (± 0.0597)	0.7402 (± 0.0143)	0.5514 (± 0.0597)	0.5621 (± 0.0489)
EntropySamplingDropout	0.5207 (± 0.0878)	0.6912 (± 0.0354)	0.5207 (± 0.0878)	0.5328 (± 0.0908)
KCenterGreedy	0.4837 (± 0.0869)	0.6392 (± 0.0678)	0.4837 (± 0.0869)	0.4693 (± 0.1108)
KMeansSampling	0.5469 (± 0.0334)	0.7194 (± 0.0365)	0.5469 (± 0.0334)	0.5728 (± 0.0278)
LeastConfidence	0.5242 (± 0.0354)	0.7431 (± 0.0297)	0.5242 (± 0.0354)	0.5383 (± 0.0417)
LeastConfidenceDropout	0.5123 (± 0.1212)	0.7088 (± 0.0419)	0.5123 (± 0.1212)	0.5223 (± 0.1509)
MarginSampling	0.5519 (± 0.0772)	0.7283 (± 0.0281)	0.5519 (± 0.0772)	0.5606 (± 0.0867)
MarginSamplingDropout	0.5624 (± 0.0706)	0.7302 (± 0.0125)	0.5624 (± 0.0706)	0.5722 (± 0.0591)
RandomSampling	0.5562 (± 0.0549)	0.7065 (± 0.0054)	0.5562 (± 0.0549)	0.5746 (± 0.0471)
SSRAEKmeansHCSampling	0.6648 (± 0.0520)	0.7433 (± 0.0231)	0.6648 (± 0.0520)	0.6713 (± 0.0428)
VCTexKmeansHCSampling	0.6431 (± 0.0409)	0.7356 (± 0.0114)	0.6431 (± 0.0409)	0.6576 (± 0.0343)

Table 2.3: Quantitative results for $n_{query} = 10$ after 8 active learning rounds (mean $\pm$ std).

icant instability, concluding the experiment with an average F1-score of only 0.5387 (± 0.2573). Because of this, the majority of the informed DAL methods successfully leveraged the larger pool of queried samples to outperform the baseline. Among the traditional techniques, MarginSampling emerged as the most effective heuristic, achieving a strong F1-score of 0.8023 (± 0.0272).

Consistent with the previous constrained scenario, the proposed hierarchical approaches maintained their absolute superiority. SSRAEKmeansHCSampling reached the highest performance with an F1-score of 0.8301 (± 0.0298), while VCTexKmeansHCSampling achieved a highly competitive 0.8134 (± 0.0255). This demonstrates that as the query size increases, the spatio-spectral extraction coupled with balanced hierarchical clustering avoids the sampling of redundant instances more effectively than pure uncertainty or margin-based methods, translating the increased data budget into substantial gains in classification accuracy.

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.6808 (± 0.0428)	0.8162 (± 0.0336)	0.6808 (± 0.0428)	0.7045 (± 0.0406)
AdversarialDeepFool	0.7244 (± 0.0464)	0.8373 (± 0.0274)	0.7244 (± 0.0464)	0.7465 (± 0.0410)
BALDDropout	0.7387 (± 0.0171)	0.7986 (± 0.0350)	0.7387 (± 0.0171)	0.7536 (± 0.0241)
EntropySampling	0.6904 (± 0.0116)	0.8226 (± 0.0102)	0.6904 (± 0.0116)	0.7180 (± 0.0124)
EntropySamplingDropout	0.5935 (± 0.0371)	0.7247 (± 0.0457)	0.5935 (± 0.0371)	0.6131 (± 0.0330)
KCenterGreedy	0.5614 (± 0.1329)	0.6960 (± 0.1236)	0.5614 (± 0.1329)	0.5850 (± 0.1279)
KMeansSampling	0.7124 (± 0.0130)	0.8118 (± 0.0096)	0.7124 (± 0.0130)	0.7305 (± 0.0088)
LeastConfidence	0.6918 (± 0.0279)	0.8126 (± 0.0076)	0.6918 (± 0.0279)	0.7133 (± 0.0256)
LeastConfidenceDropout	0.6618 (± 0.0159)	0.7787 (± 0.0402)	0.6618 (± 0.0159)	0.6855 (± 0.0183)
MarginSampling	0.7889 (± 0.0312)	0.8566 (± 0.0134)	0.7889 (± 0.0312)	0.8023 (± 0.0272)
MarginSamplingDropout	0.7034 (± 0.0384)	0.7959 (± 0.0138)	0.7034 (± 0.0384)	0.7207 (± 0.0275)
RandomSampling	0.5632 (± 0.2045)	0.7148 (± 0.1172)	0.5632 (± 0.2045)	0.5387 (± 0.2573)
SSRAEKmeansHCSampling	0.8210 (± 0.0354)	0.8577 (± 0.0154)	0.8210 (± 0.0354)	0.8301 (± 0.0298)
VCTexKmeansHCSampling	0.8022 (± 0.0276)	0.8462 (± 0.0152)	0.8022 (± 0.0276)	0.8134 (± 0.0255)

Table 2.4: Quantitative results for $n_{query} = 50$ after 8 active learning rounds (mean $\pm$ std).

For the most comprehensive experimental setting with $n_{query} = 100$, Table 2.5 details the classification metrics across all methods. With this larger labeling budget, the RandomSampling baseline achieved a highly competitive

F1-score of $0.8218 (\pm 0.0119)$ after 8 rounds. Despite this strong random baseline, several advanced active learning strategies successfully surpassed it. Among the traditional techniques, EntropySampling proved to be the most robust, achieving an F1-score of $0.8652 (\pm 0.0243)$. The proposed hierarchical methods, conversely, once again demonstrated their capacity to maximize sample utility, with SSRAEKmeansHCSampling yielding the highest overall F1-score of $0.8766 (\pm 0.0208)$, closely followed by VCTexKmeansHCSampling at $0.8712 (\pm 0.0146)$.

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.8200 (± 0.0216)	0.8858 (± 0.0120)	0.8200 (± 0.0216)	0.8356 (± 0.0192)
AdversarialDeepFool	0.8278 (± 0.0406)	0.8851 (± 0.0170)	0.8278 (± 0.0406)	0.8392 (± 0.0365)
BALDDropout	0.7942 (± 0.0168)	0.8455 (± 0.0022)	0.7942 (± 0.0168)	0.8057 (± 0.0125)
EntropySampling	0.8580 (± 0.0262)	0.8888 (± 0.0176)	0.8580 (± 0.0262)	0.8652 (± 0.0243)
EntropySamplingDropout	0.6853 (± 0.0118)	0.7906 (± 0.0434)	0.6853 (± 0.0118)	0.7069 (± 0.0141)
KCenterGreedy	0.7233 (± 0.1251)	0.7606 (± 0.1511)	0.7233 (± 0.1251)	0.7225 (± 0.1462)
KMeansSampling	0.7673 (± 0.0254)	0.8463 (± 0.0095)	0.7673 (± 0.0254)	0.7791 (± 0.0237)
LeastConfidence	0.8333 (± 0.0709)	0.8853 (± 0.0336)	0.8333 (± 0.0709)	0.8404 (± 0.0690)
LeastConfidenceDropout	0.7654 (± 0.0443)	0.8389 (± 0.0180)	0.7654 (± 0.0443)	0.7836 (± 0.0377)
MarginSampling	0.8243 (± 0.0395)	0.8894 (± 0.0155)	0.8243 (± 0.0395)	0.8390 (± 0.0344)
MarginSamplingDropout	0.7572 (± 0.0525)	0.8364 (± 0.0113)	0.7572 (± 0.0525)	0.7727 (± 0.0477)
RandomSampling	0.8150 (± 0.0106)	0.8570 (± 0.0107)	0.8150 (± 0.0106)	0.8218 (± 0.0119)
SSRAEKmeansHCSampling	0.8688 (± 0.0237)	0.8983 (± 0.0119)	0.8688 (± 0.0237)	0.8766 (± 0.0208)
VCTexKmeansHCSampling	0.8647 (± 0.0144)	0.8853 (± 0.0168)	0.8647 (± 0.0144)	0.8712 (± 0.0146)

Table 2.5: Quantitative results for $n_query = 100$ after 8 active learning rounds (mean $\pm$ std).

These quantitative findings definitively demonstrate that intelligently guided active learning can nearly match the theoretical upper-bound performance while drastically reducing labeling requirements. This exceptional data efficiency strongly aligns with the comprehensive findings of Ren et al. [2021], who established that well-calibrated DAL frameworks can consistently approximate fully supervised performance levels using only a marginal fraction of the annotation budget. Although standard DAL heuristics eventually outpaced RandomSampling, the proposed spatio-spectral hierarchical methods established a consistently higher standard of data efficiency. As recently corroborated by studies in precision agriculture [van Marrewijk et al., 2024], standard uncertainty heuristics often plateau because they fail to capture the complex structural diversity of dense crop environments. By ensuring a conceptually balanced selection through recursive clustering, our proposed methods effectively overcome this limitation, maximizing the informational utility of every queried sample.

To contextualize this impact, by applying the SSRAEKmeansHCSampling strategy with $n_query = 100$, the model achieved a remarkable F1-score of $0.8766 (\pm 0.0208)$ relying on merely 900 manually annotated images. In stark contrast, the performance upper bound of $0.9134 (\pm 0.0084)$ necessitated the full supervised labeling of all 8,086 images. This means the proposed framework experienced a minimal performance trade-off, a decrease of approximately

0.0368 units in F1-score, representing just a 4.03% reduction relative to the absolute maximum. Simultaneously, the volume of required labeled data was reduced by an astonishing 88.9%. This highlights that the proposed hierarchical sampling strategy can deliver near-optimal generalization in complex precision agriculture tasks while solving the core bottleneck of manual annotation costs.

2.4.3 Weed Classification

This subsection analyzes the classification performance at a class-specific level by examining the confusion matrices of the top-performing DAL strategies configured with $n_{query} = 100$. Figure 2.9 presents the confusion matrices for the four most effective methods SSRAEKmeansHCSampling, VCTexKmeansHCSampling, EntropySampling, and LeastConfidence, obtained after the conclusion of 8 active learning rounds (resulting in a final training subset of 900 labeled images). To establish a comprehensive analytical baseline, Figure 2.10 additionally displays the confusion matrix of the theoretical upper-bound experiment, in which the model was trained on the entire data pool of 8,086 images, providing a clear performance ceiling for comparative assessment.

SSRAEKmeansHCSampling: Figure 2.9 item a) shows the confusion matrix for the proposed SSRAEKmeansHCSampling method. This strategy demonstrated exceptional capability in correctly identifying the dominant class, *Gramineae spp.*, with 813 accurate predictions, as well as a strong performance for *Brachiaria spp.* (304 correctly predicted samples). While a moderate level of confusion remains between grass-like species, such as 76 instances of *Gramineae spp.* misclassified as *Panicum maximum*, the overall structural integrity of the predictions is robust. The method also effectively isolated *Ricinus communis* (413 correct samples) and *Broadleaf species* (129 correct samples) with minimal severe misclassifications, confirming the discriminative power of its spatio-spectral features.

VCTexKmeansHCSampling: The confusion matrix for the VCTexKmeansHCSampling approach is illustrated in Figure 2.9 item b). This method achieved the highest true-positive count for the majority class, accurately classifying 850 instances of *Gramineae spp.*. It also maintained high accuracy for *Brachiaria spp.* (287 samples) and *Ricinus communis* (403 samples). Similar to SSRAE, the primary source of error lies in the inter-class similarity of grass weeds, with 61 samples of *Gramineae spp.* being confused with *Panicum maximum*, and 40 *Brachiaria spp.* instances misclassified as *Panicum*

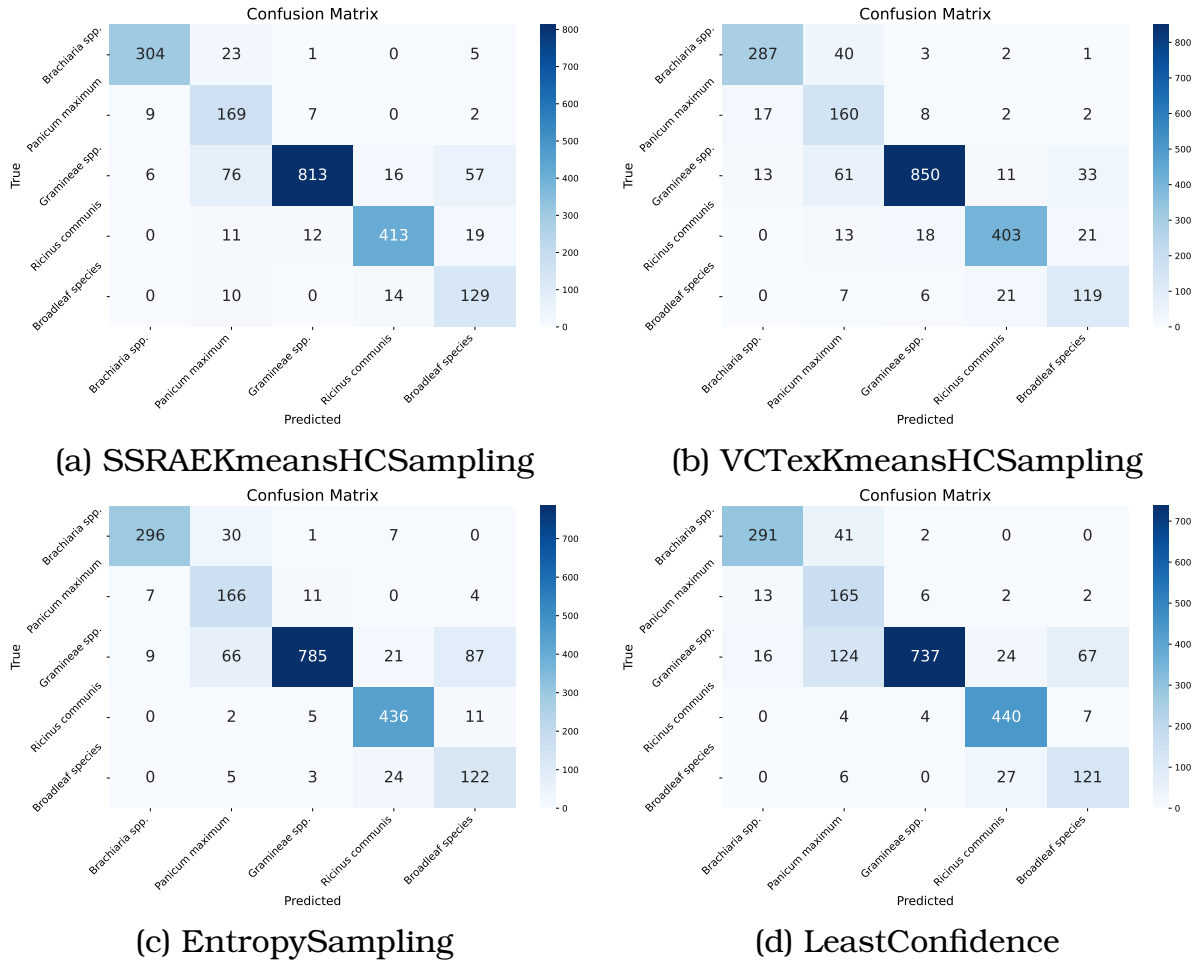


Figure 2.9: Confusion matrices of the top four DAL methods with $n_{query} = 100$: (a) SSRAEKmeansHCSampling, (b) VCTexKmeansHCSampling, (c) EntropySampling, and (d) LeastConfidence.

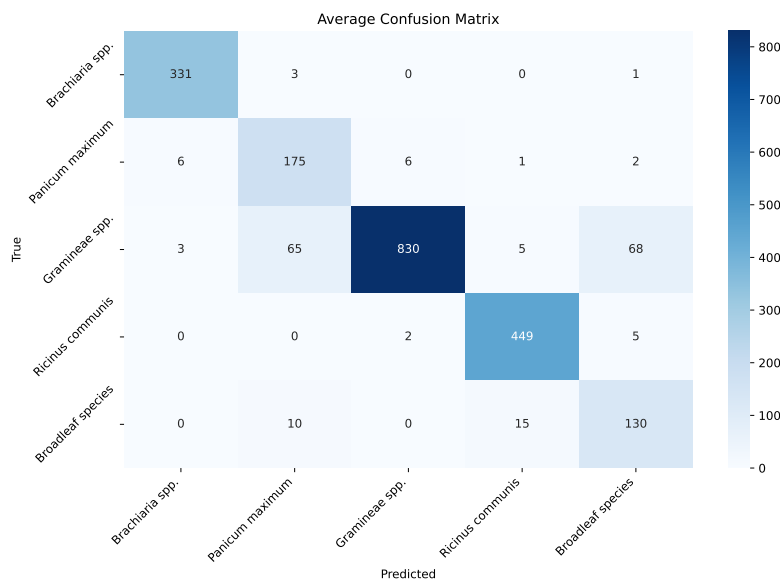


Figure 2.10: Confusion matrix for the Upper-bound experiment (8,086 images).

maximum. Despite these errors, the volumetric extraction paired with hierarchical sampling effectively stabilized the model’s decision boundaries across both majority and minority classes.

EntropySampling: Figure 2.9 item c) presents the confusion matrix for *EntropySampling*, the strongest traditional baseline. This method performed particularly well on *Ricinus communis*, achieving 436 correct classifications. *Brachiaria spp.* and *Panicum maximum* were also accurately predicted in 296 and 166 instances, respectively. Its limitations, however, become apparent when handling the dominant *Gramineae spp.* class; although 785 samples were correctly identified, the model noticeably struggled by misclassifying 87 instances as *Broadleaf species* and 66 as *Panicum maximum*, indicating that pure uncertainty sampling struggles to resolve complex boundaries in highly imbalanced domains.

LeastConfidence: The confusion matrix for *LeastConfidence* in Figure 2.9 item d) reveals significant performance degradation compared to the top methods. While it correctly identified 317 instances of *Ricinus communis*, the method demonstrated severe confusion across the remaining classes. Notably, the true-positive count for *Gramineae spp.* dropped to 608, with a massive 127 samples misclassified as *Panicum maximum* and 141 misclassified as *Broadleaf species*. In addition, 54 *Brachiaria spp.* samples were erroneously labeled as *Panicum maximum*. These extensive misclassifications highlight the vulnerability of basic uncertainty heuristics when exposed to morphologically similar weed species without the structural support of balanced data sampling.

Upper-bound Experiment (8,086 Images): Figure 2.10 illustrates the confusion matrix resulting from the "upper-bound" experiment, in which the classifier was trained using the complete dataset containing all 8,086 Pool images. The classifier achieved an overall f1-score of 0.9134 (± 0.0084), indicating high effectiveness. The diagonal elements of the matrix represent correct classifications (true positives), with notably strong results for *Gramineae spp.* (775 correctly classified instances) and *Ricinus communis* (443 correctly classified instances). Nonetheless, the classifier showed some confusion between similar categories, particularly between *Gramineae spp.* and *Panicum maximum*, where 124 instances of *Gramineae spp.* were incorrectly predicted as *Panicum maximum*, highlighting morphological similarities between these groups.

Misclassifications were fewer but still noteworthy for *Brachiaria spp.*, which had 35 instances incorrectly classified as *Panicum maximum*, suggesting overlapping visual features. Likewise, broadleaf species displayed a modest level of

confusion, with 16 instances misclassified as *Ricinus communis*. Despite these minor inaccuracies, the high accuracy across most categories underscores the robustness of the classifier when trained comprehensively, as demonstrated in this upper-bound scenario.

Comparative Discussion: Comparing these top four active learning strategies with the upper-bound experiment highlights the fundamental trade-off between manual labeling effort and classification accuracy. While all DAL methods exhibit some degree of misclassification due to the inherent morphological similarities among weed species, the proposed *SSRAEKmeansHCSampling* and *VCTexKmeansHCSampling* demonstrate a distinct advantage. They successfully mitigate the severe class overlap, particularly between grass-like classes such as *Gramineae spp.* and *Panicum maximum*, that heavily penalized standard uncertainty heuristics like *LeastConfidence* and *EntropySampling*. Naturally, the upper-bound model, leveraging the full dataset of 8,086 images, produces cleaner decision boundaries with fewer misclassifications. Nonetheless, these comparative results conclusively confirm that the proposed spatio-spectral hierarchical approaches can approximate this theoretical ceiling remarkably well using merely 900 annotated samples. This data efficiency underscores their vital utility for scalable, real-world precision agriculture applications.

2.4.4 Qualitative Analysis and Visual Discussion

In this section, we focus on the proposed *SSRAEKmeansHCSampling* and *VCTexKmeansHCSampling* methods with $n_{query} = 100$, as they achieved the highest overall performance among all evaluated DAL strategies. To provide a deeper understanding of the models' behavior in complex agricultural environments, we analyze qualitative examples by reviewing specific visual cases where the networks successfully classified the weed species, alongside instances where misclassifications occurred. Our visual analysis aims to elucidate the specific strengths and remaining limitations of the proposed feature extraction and hierarchical sampling pipelines, directly correlating them with the error patterns previously observed in the confusion matrices.

Figure 2.11 presents five representative examples of correct classifications, one for each class in the dataset, achieved by the proposed *SSRAEKmeansHCSampling* method. These qualitative examples highlight the model's robust capability to discriminate between complex weed categories despite severe real-world challenges, such as varying soil backgrounds, unpredictable lighting conditions, and the inherent low resolution of drone-captured orthomosaics. The correct identification of *Brachiaria spp.* (Figure 2.11a), *Panicum*

maximum (Figure 2.11b), and *Gramineae spp.* (Figure 2.11c) is particularly notable. Given the extreme visual similarity among these grass-like species, which share thin, elongated leaf structures and dense tufted growth patterns, these true positives confirm that the spatio-spectral feature extraction effectively captures subtle morphological and textural nuances. The accurate classification of *Ricinus communis* (Figure 2.11d) and *Broadleaf species* (Figure 2.11e) further demonstrates that the hierarchical sampling strategy successfully retains informative samples from morphologically distinct minority classes, preventing them from being overshadowed by the dominant grasses.

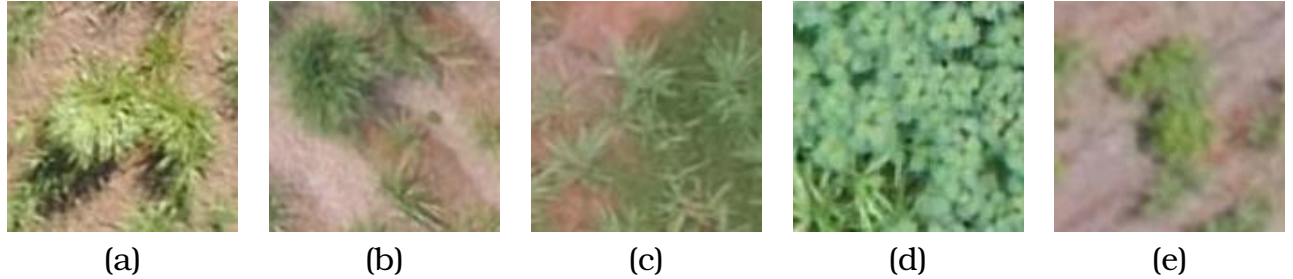


Figure 2.11: Examples of correct classifications obtained by the *SSRAEKmeansHCSampling* method with $n_{query} = 100$. (a) *Brachiaria spp.*, (b) *Panicum maximum*, (c) *Gramineae spp.*, (d) *Ricinus communis*, and (e) *Broadleaf species*.

Figure 2.12, in contrast, showcases five representative examples of misclassifications made by the *SSRAEKmeansHCSampling* method. A critical visual analysis of these error cases reveals that the model’s limitations are strongly tied to extreme morphological similarities and early plant phenological stages. As observed in Figures 2.12a, 2.12b, and 2.12c, distinguishing between *Gramineae spp.*, *Brachiaria spp.*, and *Panicum maximum* remains highly challenging. These grass-like weeds share nearly identical spectral signatures, leaf textures, and dense tufted architectures. When subjected to varying illumination, soil background interference, and the inherent limits of drone orthonomosaic resolution, the spatio-spectral encoder can occasionally blur the boundaries between these specific classes.

Figures 2.12d and 2.12e also illustrate misclassifications between *Ricinus communis* and *Broadleaf species*. These specific errors frequently occur when the plants are in early seedling stages. At this young growth phase, the distinct, highly lobed star-shaped leaves characteristic of mature *Ricinus communis* have not yet fully developed, causing their visual features to closely mimic generic broadleaf weeds. These challenging boundary cases indicate that while the proposed active learning strategy drastically improves data efficiency, resolving early-stage phenological ambiguities might require higher spatial resolution or multi-temporal image integration in future works.

Figure 2.13 similarly illustrates five representative examples of correct classifications achieved by the *VCTexKmeansHCSampling* method, covering all



Figure 2.12: Examples of incorrect classifications obtained by the proposed SSRAEKmeansHCSampling method with $n_{query} = 100$. (a) Correct: *Brachiaria spp.*, Predicted: *Panicum maximum*; (b) Correct: *Panicum maximum*, Predicted: *Brachiaria spp.*; (c) Correct: *Gramineae spp.*, Predicted: *Panicum maximum*; (d) Correct: *Ricinus communis*, Predicted: *Broadleaf species*; (e) Correct: *Broadleaf species*, Predicted: *Ricinus communis*.

dataset classes. The visual complexity of these samples, characterized by overlapping leaves, self-shadowing, and dense plant clusters, highlights the specific strengths of the volumetric color texture (VCTex) extraction. Processing the input images as 3D color-texture cubes empowers the model to successfully capture rich textural gradients and spatial pixel relationships that are crucial for differentiating highly similar species. For instance, the accurate classification of the grass-like weeds *Brachiaria spp.* (Figure 2.13a), *Panicum maximum* (Figure 2.13b), and *Gramineae spp.* (Figure 2.13c), demonstrates that the volumetric approach effectively maps the distinct tufted architectures and linear leaf patterns of these plants. Additionally, the correct identification of *Ricinus communis* (Figure 2.13d) and *Broadleaf species* (Figure 2.13e) confirms that the method accurately models the broader canopy density and distinct chromatic variations of these dicotyledonous weeds, validating the efficacy of combining 3D textural features with hierarchical sampling.

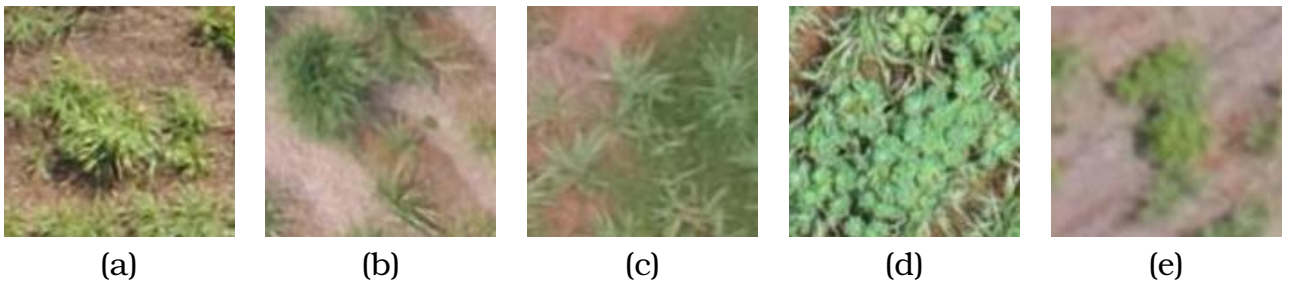


Figure 2.13: Examples of correct classifications obtained by the proposed VC-TexKmeansHCSampling method with $n_{query} = 100$. (a) *Brachiaria spp.*, (b) *Panicum maximum*, (c) *Gramineae spp.*, (d) *Ricinus communis*, and (e) *Broadleaf species*.

Figure 2.14 details five instances of misclassifications produced by the VC-TexKmeansHCSampling pipeline. Analyzing these specific limitations reveals that while volumetric texture extraction is highly robust, it remains suscepti-

ble to the most extreme cases of visual ambiguity within the dataset. As seen in Figures 2.14a, 2.14b, and 2.14c, the mutual confusion between *Brachiaria spp.*, *Panicum maximum*, and *Gramineae spp.* persists. In these specific patches, the grasses are densely intertwined or captured at an optical scale where their structural textures merge into a uniform grassy cluster, causing their 3D color-texture signatures to be virtually indistinguishable.

Following the same pattern, the errors between *Ricinus communis* and *Broadleaf species* (Figures 2.14d and 2.14e) reiterate the fundamental challenge of early phenological stages. When *Ricinus communis* lacks its mature, deeply lobed leaf geometry, either due to young age or occlusion, the textural gradient captured by the VCTex encoder closely mimics that of generic broadleaf weeds. Consequently, these visual error patterns confirm that the remaining misclassifications in both proposed hierarchical methods are driven primarily by the intrinsic optical resolution limits of the orthomosaics and the botanical similarities of the weeds, rather than inefficiencies in the active learning selection strategy.



Figure 2.14: Examples of incorrect classifications obtained by the proposed *VCTexKmeansHCSampling* method with $n_{query} = 100$. (a) Correct: *Brachiaria spp.*, Predicted: *Panicum maximum*; (b) Correct: *Panicum maximum*, Predicted: *Brachiaria spp.*; (c) Correct: *Gramineae spp.*, Predicted: *Brachiaria spp.*; (d) Correct: *Ricinus communis*, Predicted: *Broadleaf species*; (e) Correct: *Broadleaf species*, Predicted: *Ricinus communis*.

Comparing these qualitative observations with the overall quantitative metrics, it becomes evident that while the proposed hierarchical strategies, *SS-RAEKmeansHCSampling* and *VCTexKmeansHCSampling*, excel at navigating the complex feature space of agricultural images, inherent botanical ambiguities remain a formidable challenge. The successful predictions demonstrate that advanced spatio-spectral and volumetric feature extraction is highly effective at isolating subtle morphological and textural cues, even in cluttered backgrounds and varying illumination. The misclassification examples, conversely, highlight that the residual errors are largely confined to extreme boundary cases, such as highly overlapping grass tufts or early-stage dicotyledons lacking mature identifying features. Overall, these visual insights validate the robustness and data efficiency of the proposed active learning frameworks for

real-world precision agriculture. They also suggest that future improvements in classification accuracy may rely less on refining the sampling heuristics and more on integrating higher-resolution sensors or multi-temporal imagery to resolve these fine-grained biological overlaps definitively.

2.5 Discussion

2.5.1 General Discussion

The experimental results definitively demonstrate that DAL methods can effectively approach upper-bound classification performance utilizing considerably fewer labeled samples than a fully supervised approach. Quantitative metrics, such as the F1-score, alongside qualitative analyses of confusion matrices and visual predictions, reveal that the proposed hierarchical strategies, *SSRAEKmeansHCSampling* and *VCTexKmeansHCSampling*, consistently outperform traditional uncertainty-based heuristics. Notably, *SSRAEKmeansHCSampling* under the $n_{query} = 100$ configuration achieved an F1-score of 0.8766 relying on merely 900 manually annotated images, remarkably approximating the 0.9134 (± 0.0084) theoretical upper bound obtained with the entire dataset of 8,086 images.

The progressive improvement observed across active learning rounds highlights the fundamental effectiveness of selective sampling in identifying the most valuable data points. Experiments across varying query sizes ($n_{query} = 10, 50$, and 100) indicate that while larger queries naturally accelerate performance gains, the proposed methods yielded substantial improvements over random sampling even under severe budget constraints. This robustness and adaptability to different labeling budgets highlight the scalability of the proposed DAL pipelines for real-world scenarios where expert annotation is scarce and expensive [Zhang et al., 2024b].

Qualitative visual analyses of correct and misclassified images confirm that while the models excel at identifying distinct visual features, the primary remaining challenge lies in resolving subtle inter-class phenological differences and extreme morphological similarities (such as overlapping grass species). These detailed observations deepen our understanding of the models' decision boundaries and indicate that the proposed feature extraction methods effectively push the network to the physical limits of the drone sensor's optical resolution.

The superior performance of the proposed spatio-spectral and volumetric hierarchical sampling methods can be directly attributed to their ability to overcome the critical flaws of standard uncertainty sampling. Traditional heuristics like *LeastConfidence*, *EntropySampling*, and *MarginSampling* prior-

itize instances that are strictly the most ambiguous to the model. However, in highly imbalanced agricultural datasets containing visually similar classes, these methods often suffer from sampling redundancy, repeatedly selecting nearly identical confusing samples from the dominant classes while completely overlooking informative examples from minority classes. In contrast, coupling advanced latent feature extraction (SSRAE and VCTex) with balanced hierarchical clustering (*KmeansHCSampling*) ensures that the queried batches in the proposed frameworks are not only informative but also structurally diverse. This clustering mechanism prevents the active learning loop from collapsing into ambiguous, narrow decision boundaries, actively forcing the model to learn more discriminative, globally representative features. This targeted diversity fundamentally leads to superior generalization and performance much closer to the fully supervised upper bound [Gul et al., 2024].

2.5.2 Implications for Computational Approaches in Agriculture

The exceptional performance of the proposed hierarchical DAL frameworks presents profound implications for computational agriculture. The demonstrated ability of the SSRAE and VCTex pipelines, combined with K-means hierarchical sampling, to approximate the theoretical upper-bound performance using a mere fraction of the dataset fundamentally addresses one of the highest barriers in agricultural deep learning: the prohibitive cost of expert, pixel-level annotation [Kamilaris and Prenafeta-Boldú, 2018; Liakos et al., 2018]. Intelligently querying only the most structurally diverse and informative weed samples rather than redundantly sampling the majority class drastically reduces human labor and accelerates the rapid deployment of automated, highly accurate Site-Specific Weed Management (SSWM) systems.

Beyond this, the models' proven capacity to navigate severe class imbalances and extreme morphological similarities, such as differentiating between dense, tufted grasses like *Brachiaria spp.* and *Panicum maximum*, is highly relevant for real-world agronomy. In field scenarios where timely and species-specific herbicide application is critical to preventing crop yield loss, the proposed methods offer a pathway to highly reliable mapping. Integrating these advanced spatial, spectral, and volumetric extraction pipelines with drone-based orthomosaics and real-time edge processing can revolutionize how early-stage infestations are monitored and treated. Rapid interventions enabled by this capability significantly minimize blanket chemical usage, thereby lowering operational costs and reducing environmental impact [Mohanty et al., 2016].

These advancements underscore the transformative potential of shifting from generic uncertainty heuristics to domain-aware, feature-rich active learning in agricultural systems. Optimizing the data annotation bottleneck with-

out sacrificing the predictive power required to identify visually ambiguous weed species makes the proposed hierarchical DAL strategies a vital, scalable step toward fully autonomous and sustainable digital agriculture.

2.6 Conclusion

This study demonstrates that the proposed hierarchical Deep Active Learning (DAL) frameworks can achieve near-optimal classification performance in highly imbalanced weed identification tasks while drastically reducing the need for extensive manual annotation. Employing domain-aware feature extraction coupled with balanced clustering, specifically the *SSRAEKmeansHC-Sampling* strategy with $n_{query} = 100$, enabled our experiments to achieve an F1-score of 0.8766 utilizing merely 900 labeled images. This remarkable data efficiency closely approximates the 0.9134 (± 0.0084) performance upper-bound reached through full supervision on the entire dataset of 8,086 images, representing nearly an 89% reduction in required labeling effort.

The comprehensive quantitative analyses, supported by detailed confusion matrices and qualitative visual evaluations, confirm that these proposed strategies successfully mitigate the sampling redundancy that typically plagues traditional uncertainty heuristics. Strategically capturing spatio-spectral and volumetric textures, and subsequently forcing structural diversity through hierarchical sampling, allows the models to effectively navigate extreme morphological similarities among grass-like species and ambiguous early-stage dicotyledons.

Our experiments additionally revealed that even under severely constrained labeling budgets (e.g., $n_{query} = 10$ and 50), the proposed hierarchical DAL methods progressively and stably improved model performance. These results unequivocally suggest that by intelligently querying structurally diverse and informative batches rather than merely the most uncertain samples, it is possible to attain robust generalization from the earliest active learning rounds. As a result, this scalable approach substantially minimizes annotation costs, directly expediting the real-world deployment of highly accurate, automated SSWM systems in modern precision agriculture.

2.7 Limitations

Despite the significant improvements in data efficiency and classification accuracy achieved by the proposed hierarchical DAL frameworks, several inherent limitations must be acknowledged. First, while the dataset accurately reflects real-world agricultural conditions, the models' performance is fun-

damentally constrained by the specific Ground Sample Distance (GSD) and optical resolution of the drone-captured orthomosaics. As evidenced in the qualitative analysis, even the advanced spatio-spectral and volumetric encoders occasionally struggle with extreme morphological similarities. Specifically, distinguishing between intricately overlapping grass-like species (e.g., *Gramineae spp.* and *Panicum maximum*) and identifying dicotyledonous weeds in their early phenological stages (e.g., *Ricinus communis* seedlings) remain formidable challenges. These boundary-case ambiguities are driven primarily by biological realities and physical sensor limitations rather than inefficiencies in the active learning sampling heuristics.

Another limitation is that the current study was conducted on a static, curated dataset pool. While this environment is standard for benchmarking algorithmic data efficiency, it may not fully capture the continuous, dynamic distribution shifts (e.g., varying soil moisture, seasonal lighting changes, and evolving plant growth stages) encountered in broader, longitudinal agricultural contexts [Settles, 2009; Oliver et al., 2018]. Finally, while the proposed feature extraction pipelines are robust, their computational overhead compared to simpler heuristics must be carefully managed. Exploring real-time, on-edge active learning scenarios in dynamic agricultural settings will be crucial for translating these empirical findings into practical, scalable, and instantaneous solutions for precision agriculture [Ranganathan et al., 2017].

2.8 Future Research Directions

Building upon the robust foundation established by the SSRAE and VCTex hierarchical active learning frameworks, several promising avenues for future research emerge. First, to overcome the physical limitations of RGB optical resolution and the ambiguities of early phenological stages, future studies should extend these DAL methods to multi-temporal and multi-sensor agricultural datasets. Incorporating temporal image sequences, tracking plant growth over time, or fusing hyperspectral and LiDAR data could provide the definitive features needed to separate morphologically identical grass-like weeds under varying field conditions [Kamilaris and Prenafeta-Boldú, 2018; Liakos et al., 2018].

Transitioning the proposed active learning pipelines from image-level classification to dense pixel-level tasks, such as semantic or instance segmentation, represents a critical next step. Because pixel-wise annotation is exponentially more expensive and time-consuming than image classification, the data efficiency demonstrated by our hierarchical sampling strategies would be even more transformative for training precise, segmentation-based Site-

Specific Weed Management (SSWM) models.

Additionally, exploring multimodal DAL architectures offers a highly viable path for significant algorithmic improvements. In an agricultural context, this multimodal integration would involve fusing the spatio-spectral visual features with rich agronomic metadata (e.g., soil composition, weather conditions, and days after sowing) serving as contextual textual/tabular priors to guide the active sampling process [Yang et al., 2024; Bang et al., 2024].

Finally, there is a critical industrial need for the development of real-time, adaptive, active learning frameworks capable of operating on edge devices, such as UAVs or agricultural machinery, in dynamic environments [Ren et al., 2021]. Enabling these models to continuously adapt to new distribution shifts in the field, querying only the most informative structural anomalies on the fly, will fully realize the potential of computational approaches. Ultimately, advancing these targeted, domain-aware DAL techniques is essential to driving the digital transformation of agriculture, drastically reducing annotation bottlenecks, and ensuring sustainable crop management without any compromise in predictive performance.

Conclusions and Future Work

3.1 *Conclusions*

To date, research efforts have focused on the study of DAL techniques and the acquisition of the target dataset. Following the preparation of the dataset and the development of a robust coding infrastructure, empirical experiments were carried out, and a scientific paper was drafted, subsequently incorporated as Chapter 2. The principal objective of this phase was to implement and assess various DAL methods concerning the proposed task of classifying weed images. As demonstrated in Chapter 2, these methods exhibited strong performance, thereby highlighting the significant potential of DAL approaches to address the challenges inherent in our classification problem. In the second stage of the work, we intend to add another deep learning task and explore the efficiency of DAL methods in semantic image segmentation problems.

Recent studies emphasize that semantic segmentation demands extensive, costly, and time-consuming pixel-level manual annotations [Yujian et al., 2022; Chakravarthy et al., 2022; Csurka et al., 2023]. Zhang et al. [Zhang et al., 2025] introduce active learning techniques to efficiently train semantic segmentation models with limited annotated data, thereby alleviating the burden of ground-truth labeling. Reza et al. [Reza et al., 2025] highlight that obtaining large-scale, point-by-point manual annotations remains a significant challenge and propose a region-based active learning strategy. Furthermore, the prohibitive cost of obtaining dense pixel-level annotations for supervised training is identified as a central obstacle in semantic segmentation [Gao et al., 2023]. Preparing datasets for training segmentation deep neural network is

inherently laborious, time-intensive, and expensive, as it necessitates both dense pixel-wise annotations and large volumes of labeled images to establish reliable ground truth and support large-scale unsupervised segmentation [Zhang et al., 2025; Reza et al., 2025; Gao et al., 2023; Luo et al., 2018].

Encouraged by the promising classification results achieved with conventional DAL techniques, the next phase of this work will explore more advanced deep active learning frameworks, particularly those that leverage multimodal models [Yang et al., 2024; Bang et al., 2024]. To this end, we will expand our current dataset to encompass a broader range of scenarios and rigorously test the limits of these novel models on larger data collections. Furthermore, recognizing the substantial annotation effort required for pixel-wise segmentation [Yujian et al., 2022], we will investigate the integration of specialized active learning strategies for segmentation, which have shown potential to significantly lower labeling costs and accelerate result generation [Mittal et al., 2024]. In this context, we aim to develop and apply multimodal DAL methods for both classification and semantic segmentation tasks, thereby enhancing the scope and robustness of our proposed solutions.

3.2 *Future works*

Integrating visual and textual data has shown promise in enhancing model performance in various computer vision tasks. Building upon the methodologies presented by Yang et al. [Yang et al., 2024] and Bang et al. [Bang et al., 2024], we propose a framework that combines active learning strategies with vision-language models to improve the efficiency and accuracy of agriculture precision systems.

- **Multimodal Dataset Collection:** Gather a diverse set of weed plant images across various growth stages, lighting conditions, and backgrounds, and pair each image with detailed textual descriptions (morphological characteristics, habitat information, etc.) to create a rich, complementary dataset.
- **Vision-Language Model Adaptation:** Employ a pre-trained language-vision model (e.g., CLIP) that encodes images and text into a shared embedding space and apply fast learning techniques to specialize in semantic weed species classification and segmentation tasks [Bang et al., 2024].
- **Active Learning Sample Selection:** Use the Plug and Play Active Learning (PPAL) strategy to combine uncertainty-based and diversity-based sampling methods, selecting the most informative multimodal samples

for annotation to maximize performance gains per annotation cost [Yang et al., 2024].

- **Iterative Training and Refinement:** Fine-tune the adapted vision–language model on the selected samples (image + text), evaluate using accuracy, precision, recall, and F1-score, and iteratively update the model with new annotations-monitoring performance improvements to guide further sample selection.

3.3 Execution cronogram

During the first and second years of the doctoral program, all credits required for the qualification examination were completed. Table 3.1 lists the courses, seminars, and other academic activities undertaken during this period. The article developed in chapter 2 has been submitted to the journal **IEEE Geoscience and Remote Sensing Letters** and is currently being evaluated by the editors and reviewers¹.

Year-Semester	Course	Grade
2023-1	Analysis of Algorithms	C
2023-1	Teaching Internship II	–
2023-1	Artificial Intelligence	A
2023-2	Computer Architecture	A
2023-2	Teaching Internship III	–
2023-2	Directed Study in Methodology and Techniques of Computing	A
2023-2	Computing Seminars	A
2023-2	Special Topics	A
2024-1	Validation of Master’s Degree Course credits	–
2024-1	Submission of an article to a journal	–

Table 3.1: Record of the courses and activities carried out within the program’s framework in the first two years of the doctoral course.

To achieve the objectives outlined in Section 1.4, it is essential to implement the steps proposed in Section 3.2 over the next two years, the deadline set for the completion of this study, as detailed in the biannual schedule presented in Table 3.2.

¹<https://sibgrapi.sbc.org.br/2025/index.php/grsl/>

* Activity/Stage	2025		2026	
	1º Sem.	2º Sem.	1º Sem.	2º Sem.
Qualification Exam and Revisions	•			
Dataset Expansion (Classification)	•	•		
Dataset Creation (Segmentation)	•	•		
Multimodal DAL Experiments		•	•	
Performance Evaluation and Refinement			•	
Manuscript Preparation (Classification)			•	
Manuscript Preparation (Segmentation)			•	
Final Thesis Writing				•

Table 3.2: Project development timeline leading up to the Thesis Defense.

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